

The Coefficient Algebra of Exponentials of Rational Functions

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July 3, 2026

Abstract

Let R be a rational function with a pole of order m at a and principal part $\Pi(w) = \sum_{j=1}^m c_j w^{-j}$, $w = z - a$. The Laurent coefficients of e^R on a punctured disk about a are governed by the coefficients Λ_N of $\exp(\Pi)$, which satisfy the first-order recurrence $N\Lambda_N = \sum_{j=1}^{\min(m,N)} j c_j \Lambda_{N-j}$. We take this recurrence as the primitive object, construct from it a graded commutative Hopf algebra $\mathfrak{B}_m = \mathbb{C}[c_1, \dots, c_m]$ carrying the sequence (Λ_N) , and prove its structure theory: $\mathfrak{B}_m$ is canonically the quotient of the Hopf algebra of symmetric functions by the Hopf ideal $(p_{m+1}, p_{m+2}, \dots)$, under which Λ_N corresponds to the complete homogeneous symmetric function h_N , c_j to p_j/j , and the universal recurrence to Newton's identity. The sequence (Λ_N) is a divided-power sequence; as functions of c_1 the $N!\Lambda_N$ form an Appell sequence; the coefficient matrix is an element of the Appell subgroup of the exponential Riordan group; $N!\Lambda_N$ enumerates set partitions with block sizes at most m ; the graded Hopf automorphism group is the torus $(\mathbb{C}^\times)^m$; and the Hankel determinants of (Λ_N) are, up to explicit sign, rectangular Schur functions. We then prove that the analytic Laurent-coefficient construction is a model of the algebra, with quantitative growth bounds; we identify the classical specializations exactly (modified Bessel functions for $m = 1$, Hermite-type polynomials for $m = 2$, Wright generalized Bessel functions for single-term principal parts against a linear entire exponent) and refrain from claiming reductions that do not exist for $m \geq 3$; we prove holonomy of the full Laurent coefficient sequence at every pole, with an explicit recurrence and order bound derived from $y' = R'y$; and we prove an initiality theorem and a recognition theorem that together constitute the axiomatic characterization, with independence of the hypotheses established by counterexamples. The final section develops, from the fractional Riccati equation itself (Müntz–Tau series, moment functional, the polynomial Chebyshev–Wheeler three-term recurrence, its diagonal Padé numerator and denominator, the Jacobi-matrix spectrum, and real-ray Stahl convergence, all proved in full), the rough Heston characteristic function $\varphi_M = \exp(t A_M(x)/B_M(x))$, $x = t^\alpha$, as the exponential of a diagonal Padé approximant in the coefficient algebra: we give the exact local Laurent data of φ_M at the zeros of B_M , prove convergence of Padé prices to exact prices under explicit hypotheses on a fixed Lewis-type contour with no contour deformation through approximant poles, and derive and prove exact European call and put price formulas directly from the cumulant generating function via Gil–Pelaez inversion, including a certified real-axis pole-freeness result for the Padé approximant's evaluation contour. Statements that remain open (the domination hypothesis feeding price convergence, resurgent/alien-derivative structure, uniform-in-parameter domination, exponential-motive positioning) are listed as open problems, not theorems.

1 The universal recurrence

Throughout, $m \geq 1$ is a fixed integer and $c_1, \dots, c_m$ are independent commuting indeterminates over $\mathbb{C}$. Set

$$\mathfrak{B}_m := \mathbb{C}[c_1, \dots, c_m],$$

graded by $\text{wt}(c_j) = j$, so that $\mathfrak{B}_m = \bigoplus_{N \geq 0} (\mathfrak{B}_m)_N$ with $(\mathfrak{B}_m)_0 = \mathbb{C}$. All constructions in Sections 1–8 are exact identities in $\mathfrak{B}_m$ or in the formal power series ring $\mathfrak{B}_m[[v]]$; no analysis enters until Section 5.

Definition 1 $E(v) := \exp\left(\sum_{j=1}^m c_j v^j\right) \in \mathfrak{B}_m[[v]]$, and $\Lambda_N \in \mathfrak{B}_m$ is defined by $E(v) = \sum_{N \geq 0} \Lambda_N v^N$. We set $\Lambda_N := 0$ for $N < 0$.

The exponential is well defined because $\sum_j c_j v^j$ has zero constant term. Under the substitution $v = w^{-1}$, E becomes the formal expansion of $\exp\left(\sum_{j=1}^m c_j w^{-j}\right)$, the exponential of the principal part of a rational function at a pole of order $\leq m$; the analytic counterpart is Theorem 10.

Lemma 1 (Universal recurrence) $\Lambda_0 = 1$ and, for all $N \geq 1$,

$$N \Lambda_N = \sum_{j=1}^{\min(m, N)} j c_j \Lambda_{N-j}. \quad (1)$$

Proof E satisfies the linear first-order equation $E'(v) = \left(\sum_{j=1}^m j c_j v^{j-1}\right) E(v)$ in $\mathfrak{B}_m[[v]]$, obtained by formal differentiation of the exponential. The coefficient of v^{N-1} on the left is $N \Lambda_N$; on the right it is $\sum_{j=1}^m j c_j \Lambda_{N-j}$, in which terms with $j > N$ vanish by the convention $\Lambda_{<0} = 0$. The initial value $\Lambda_0 = E(0) = 1$. $\square$

Lemma 2 (Partition formula) For all $N \geq 0$,

$$\Lambda_N = \sum_{\substack{k_1, \dots, k_m \geq 0 \\ \sum_j j k_j = N}} \prod_{j=1}^m \frac{c_j^{k_j}}{k_j!}. \quad (2)$$

In particular Λ_N is homogeneous of weight N .

Proof $E(v) = \prod_{j=1}^m \exp(c_j v^j) = \prod_{j=1}^m \sum_{k_j \geq 0} \frac{c_j^{k_j}}{k_j!} v^{j k_j}$; the product of the m series is computed by collecting the coefficient of v^N , which is (2). Homogeneity is immediate since the monomial $\prod c_j^{k_j}$ has weight $\sum j k_j = N$. $\square$

Proposition 1 (Bell-polynomial form) $N! \Lambda_N = B_N(1! c_1, 2! c_2, \dots, m! c_m, 0, \dots, 0)$, where B_N is the complete Bell polynomial, defined by $\exp\left(\sum_{j \geq 1} x_j t^j / j!\right) = \sum_{N \geq 0} B_N(x_1, \dots, x_N) t^N / N!$ (Comtet [5, §3.3]).

Proof Substitute $x_j = j! c_j$ for $j \leq m$ and $x_j = 0$ for $j > m$ in the defining generating identity of B_N ; the left side becomes $E(t)$. $\square$

Lemma 3 (Derivations) For $1 \leq j \leq m$ and all N ,

$$\frac{\partial \Lambda_N}{\partial c_j} = \Lambda_{N-j}.$$

Proof $\partial_{c_j} E(v) = v^j E(v)$; compare coefficients of v^N . $\square$

Remark 1 Lemma 1 is exactly the Euler homogeneity relation: the grading operator $\mathcal{E} = \sum_{j=1}^m j c_j \partial_{c_j}$ satisfies $\mathcal{E} \Lambda_N = \sum_j j c_j \Lambda_{N-j}$ by Lemma 3, and $\mathcal{E} \Lambda_N = N \Lambda_N$ by Lemma 2. Conversely, any sequence of weight-homogeneous elements satisfying Lemma 3 with $\Lambda_0 = 1$ satisfies (1). The universal recurrence, the lowering operators, and the grading are one structure viewed three ways.

Theorem 1 (Generation and freeness) $\mathbb{C}[\Lambda_1, \dots, \Lambda_m] = \mathfrak{B}_m$, and $\Lambda_1, \dots, \Lambda_m$ are algebraically independent over $\mathbb{C}$. Consequently $(\Lambda_N)_{N \geq 0} \subset \mathbb{C}[\Lambda_1, \dots, \Lambda_m]$: the entire coefficient sequence lives in the polynomial algebra on its first m terms.

Proof For $1 \leq N \leq m$, the multi-index $k_N = 1$, $k_j = 0$ ($j \neq N$) is the unique solution of $\sum j k_j = N$ using the variable c_N ; every other solution has $k_N = 0$ and involves only $c_1, \dots, c_{N-1}$. Hence $\Lambda_N = c_N + q_N(c_1, \dots, c_{N-1})$ with q_N polynomial. By induction, $c_N \in \mathbb{C}[\Lambda_1, \dots, \Lambda_N]$, so $\mathbb{C}[\Lambda_1, \dots, \Lambda_m] \supseteq \mathbb{C}[c_1, \dots, c_m]$; the reverse inclusion is trivial. Since $\text{trdeg}_{\mathbb{C}} \mathfrak{B}_m = m$ and $\Lambda_1, \dots, \Lambda_m$ generate it, they are algebraically independent. For $N > m$, $\Lambda_N \in \mathfrak{B}_m = \mathbb{C}[\Lambda_1, \dots, \Lambda_m]$; explicitly, iterate (1). $\square$

2 Hopf structure and the symmetric-function realization

Definition 2 Equip $\mathfrak{B}_m$ with the unique bialgebra structure for which each c_j is primitive: $\Delta c_j = c_j \otimes 1 + 1 \otimes c_j$, $\varepsilon(c_j) = 0$, extended as algebra morphisms. Since $\mathfrak{B}_m$ is graded and connected ($(\mathfrak{B}_m)_0 = \mathbb{C}$), it is automatically a Hopf algebra, with antipode determined by $S(c_j) = -c_j$ (graded connected bialgebras possess antipodes; see [20, §1]).

Theorem 2 (Divided-power property) $\Delta \Lambda_N = \sum_{i=0}^N \Lambda_i \otimes \Lambda_{N-i}$ for all $N \geq 0$, $\varepsilon(\Lambda_N) = \delta_{N,0}$, and $\sum_{i=0}^N \Lambda_i S(\Lambda_{N-i}) = \delta_{N,0}$, with $S(\Lambda_N) = \Lambda_N(-c_1, \dots, -c_m)$.

Proof Apply the algebra morphism Δ coefficient-wise in $(\mathfrak{B}_m \otimes \mathfrak{B}_m)[[v]]$:

$$\sum_N \Delta \Lambda_N v^N = \Delta E(v) = \exp \left(\sum_j (c_j \otimes 1 + 1 \otimes c_j) v^j \right) = \exp \left(\sum_j (c_j \otimes 1) v^j \right) \exp \left(\sum_j (1 \otimes c_j) v^j \right),$$

the factorization being valid because the two exponents commute. The right side equals $(\sum_i \Lambda_i \otimes 1 v^i) (\sum_k 1 \otimes \Lambda_k v^k)$; compare coefficients. The counit and antipode identities follow the same way from $\varepsilon E(v) = 1$ and $SE(v) = E(v)^{-1} = \exp(-\sum c_j v^j)$. $\square$

Thus (Λ_N) is a *divided-power sequence* in the graded connected Hopf algebra $\mathfrak{B}_m$: $\Lambda_N \in (\mathfrak{B}_m)_N$, $\Lambda_0 = 1$, $\Delta \Lambda_N = \sum \Lambda_i \otimes \Lambda_{N-i}$. Equivalently, the series $E(v)$ is group-like in the completed Hopf algebra $\mathfrak{B}_m[[v]]$.

Theorem 3 (Primitives) $\text{Prim}(\mathfrak{B}_m) = \bigoplus_{j=1}^m \mathbb{C} c_j$; in particular the primitive space in weight j is one-dimensional for $1 \leq j \leq m$ and zero for $j > m$.

Proof Each c_j is primitive by construction. Conversely, let $f = \sum_{\alpha} a_{\alpha} c^{\alpha}$ be primitive, with multi-index notation $c^{\alpha} = \prod c_j^{\alpha_j}$. Then $\Delta(c^{\alpha}) = \prod_j (c_j \otimes 1 + 1 \otimes c_j)^{\alpha_j} = \sum_{0 \leq \beta \leq \alpha} \binom{\alpha}{\beta} c^{\beta} \otimes c^{\alpha-\beta}$. The component of $\Delta f - f \otimes 1 - 1 \otimes f$ in $\mathfrak{B}_m^+ \otimes \mathfrak{B}_m^+$ ($\mathfrak{B}_m^+ := \ker \varepsilon$) is $\sum_{\alpha} a_{\alpha} \sum_{0 < \beta < \alpha} \binom{\alpha}{\beta} c^{\beta} \otimes c^{\alpha-\beta}$. The elements $\{c^{\beta} \otimes c^{\gamma} : \beta, \gamma \neq 0\}$ are linearly independent in $\mathfrak{B}_m \otimes \mathfrak{B}_m$, and for fixed (β, γ) the total coefficient is $a_{\beta+\gamma} \binom{\beta+\gamma}{\beta}$. Primitivity therefore forces $a_{\alpha} = 0$ whenever α admits a decomposition $\alpha = \beta + \gamma$ with $\beta, \gamma \neq 0$, i.e. whenever $|\alpha| := \sum_j \alpha_j \geq 2$. Hence f is a linear combination of $1, c_1, \dots, c_m$, and $\varepsilon(f) = 0$ kills the constant. $\square$

2.1 The symmetric-function isomorphism

Let $\text{Sym} = \mathbb{C}[p_1, p_2, \dots]$ be the Hopf algebra of symmetric functions over $\mathbb{C}$, with power sums p_r primitive, complete homogeneous symmetric functions h_N , and Schur functions s_{λ} (Macdonald [19, Ch. I]). Write $\mathfrak{B}_{\infty} := \mathbb{C}[c_1, c_2, \dots]$ for the evident $m = \infty$ version of Definitions 1 and 2; Lemmas 1–3 and Theorems 1–3 hold verbatim with $\min(m, N)$ replaced by N .

Theorem 4 (Symmetric-function realization) The assignment $\Psi(p_j) := j c_j$ extends uniquely to an isomorphism of graded Hopf algebras $\Psi : \text{Sym} \xrightarrow{\sim} \mathfrak{B}_{\infty}$, and

$$\Psi(h_N) = \Lambda_N \quad (N \geq 0).$$

Under Ψ , Newton's identity $N h_N = \sum_{j=1}^N p_j h_{N-j}$ becomes the universal recurrence (1). Moreover the ideal $I_m := (c_{m+1}, c_{m+2}, \dots) \subset \mathfrak{B}_{\infty}$ is a graded Hopf ideal, and

$$\mathfrak{B}_m \cong \mathfrak{B}_{\infty}/I_m \cong \text{Sym}/(p_{m+1}, p_{m+2}, \dots)$$

as graded Hopf algebras, compatibly with the sequences: $h_N \mapsto \Lambda_N$ throughout.

Proof Sym and $\mathfrak{B}_{\infty}$ are free commutative algebras on primitive generators p_j , resp. c_j , one in each weight $j \geq 1$; Ψ matches them by weight with the nonzero scalar j , hence is an isomorphism of graded algebras. It is a coalgebra map because it sends primitives to primitives and both coproducts are algebra morphisms determined on generators; a bialgebra isomorphism of Hopf algebras automatically respects antipodes. For the sequence: the generating identity $\sum_{N \geq 0} h_N t^N = \exp\left(\sum_{r \geq 1} p_r t^r / r\right)$ ([19, I.(2.10)]) maps under Ψ (applied coefficient-wise in t) to $\sum_N \Psi(h_N) t^N = \exp\left(\sum_r c_r t^r\right) = \sum_N \Lambda_N t^N$, so $\Psi(h_N) = \Lambda_N$. Newton's identity is [19, I.(2.11)']; applying Ψ gives (1) (and, independently, Lemma 1 re-proves Newton's identity via the differential equation $E' = (\sum j c_j v^{j-1}) E$). For the quotient: I_m is generated by primitives, so $\Delta I_m \subseteq I_m \otimes \mathfrak{B}_{\infty} + \mathfrak{B}_{\infty} \otimes I_m$, $\varepsilon(I_m) = 0$, $S(I_m) \subseteq I_m$: a Hopf ideal. The quotient map $\mathfrak{B}_{\infty} \rightarrow \mathfrak{B}_m$, $c_j \mapsto c_j$ ($j \leq m$), $c_j \mapsto 0$ ($j > m$), carries $\Lambda_N^{(\infty)}$ to Λ_N by Lemma 2 (multi-indices with $k_j > 0$ for some $j > m$ die). $\square$

Remark 2 $\mathfrak{B}_\infty$ is *not* the Faà di Bruno Hopf algebra: the latter is the coordinate Hopf algebra of the group of formal diffeomorphisms and is not cocommutative (*Figueroa-Gracia-Bondía [9]*), whereas $\mathfrak{B}_\infty \cong \text{Sym}$ is commutative and cocommutative. The relation to Bell polynomials is the coefficient identity of Proposition 1, not an isomorphism with the Faà di Bruno object. This answers, precisely, the question “is it the Bell algebra”: the coefficient algebra is the algebra of symmetric functions, in which the Λ_N are the complete homogeneous basis re-expressed through rescaled power sums.

Remark 3 (Hall algebra) The classical Hall algebra of finite modules over a discrete valuation ring with finite residue field is isomorphic, after the standard rescaling, to Sym ([19, Ch. II–III]). Via Theorem 4, $\mathfrak{B}_\infty$ therefore *does* arise as a Hall algebra, and $\mathfrak{B}_m$ as an explicit Hopf quotient of one. We record this as an identification available by citation; we make no use of the finer (Hall–Littlewood, q -) structure here, and lifting the coefficient sequence (Λ_N) to the q -deformed level is Open Problem 4.

3 Combinatorial and matrix realizations

Theorem 5 (Species / set-partition interpretation) For $N \geq 1$,

$$N! \Lambda_N = \sum_{\substack{\pi \vdash [N] \\ \text{all blocks of size} \leq m}} \prod_{B \in \pi} |B|! c_{|B|},$$

the sum over set partitions π of $\{1, \dots, N\}$ with block sizes at most m . Equivalently, $E(v)$ is the exponential generating function of the species “sets of blocks,” each block of size $j \leq m$ weighted $j! c_j$.

Proof The number of set partitions of $[N]$ with exactly k_j blocks of size j is $N! / \prod_j ((j!)^{k_j} k_j!)$. Hence

$$\sum_{\pi} \prod_{B \in \pi} |B|! c_{|B|} = \sum_{k : \sum j k_j = N} \frac{N!}{\prod_j (j!)^{k_j} k_j!} \prod_j (j! c_j)^{k_j} = N! \sum_{k : \sum j k_j = N} \prod_j \frac{c_j^{k_j}}{k_j!} = N! \Lambda_N$$

by Lemma 2. The species statement is the exponential formula (*Flajolet–Sedgewick [10, II.2]; Bergeron–Labelle–Leroux [3]*). $\square$

Theorem 6 (Appell structure) Fix $c_2, \dots, c_m$ and regard $p_N(c_1) := N! \Lambda_N$ as a polynomial in c_1 . Then $(p_N)_{N \geq 0}$ is an Appell sequence: $p'_N(c_1) = N p_{N-1}(c_1)$, with exponential generating function $\sum_N p_N(c_1) t^N / N! = e^{c_1 t} A(t)$, $A(t) = \exp\left(\sum_{j=2}^m c_j t^j\right)$.

Proof Lemma 3 with $j = 1$ gives $\partial_{c_1} \Lambda_N = \Lambda_{N-1}$, i.e. $p'_N = N p_{N-1}$; the generating function is Definition 1 with the c_1 -factor split off. Appell sequences are exactly the Sheffer sequences with identity delta series, so (p_N) is Sheffer of Appell type (*Roman [22, §2.5]*). $\square$

Theorem 7 (Riordan-group realization) *The lower-triangular matrix $D = (d_{n,k})_{n,k \geq 0}$, $d_{n,k} := \frac{n!}{k!} \Lambda_{n-k}$, is the exponential Riordan array $(E(t), t)$, an element of the Appell subgroup of the exponential Riordan group. The group law restricted to this subgroup, $(E(t), t) \cdot (E'(t), t) = (E(t)E'(t), t)$, corresponds to addition $c_j \mapsto c_j + c'_j$ of parameters, i.e. to the coproduct of Theorem 2.*

Proof By definition (Shapiro et al. [23]; Barry [2]) the exponential Riordan array $(g(t), f(t))$ has entries $d_{n,k} = \frac{n!}{k!} [t^{n-k}] g(t) f(t)^k / t^k$ reduced, for $f(t) = t$, to $d_{n,k} = \frac{n!}{k!} [t^{n-k}] g(t)$; with $g = E$ this is $\frac{n!}{k!} \Lambda_{n-k}$. Matrix multiplication of Appell arrays multiplies the g 's; $E(t; c)E(t; c') = E(t; c + c')$ since the exponents add. The identification with the coproduct is Theorem 2 read through $\Lambda_N(c + c') = \sum_i \Lambda_i(c) \Lambda_{N-i}(c')$, which is the image of $\Delta \Lambda_N$ under evaluation at (c, c') . $\square$

3.1 Hankel determinants are rectangular Schur functions

Theorem 8 (Hankel–Schur identity) *In $\mathfrak{B}_\infty \cong \text{Sym}$ (hence, after applying the quotient of Theorem 4, as identities in $\mathfrak{B}_m$), for every $n \geq 1$:*

$$\det(\Lambda_{i+j})_{0 \leq i, j \leq n-1} = (-1)^{\binom{n}{2}} \Psi(s_{((n-1)^n)}), \quad \det(\Lambda_{i+j+1})_{0 \leq i, j \leq n-1} = (-1)^{\binom{n}{2}} \Psi(s_{(n^n)}),$$

where (r^n) denotes the $n \times r$ rectangular partition.

Proof Work in Sym with $\Lambda_N = h_N$ (Theorem 4). Reverse the rows of the Hankel matrix by $i \mapsto n - 1 - i$; this multiplies the determinant by the sign of the reversal permutation, $(-1)^{\binom{n}{2}}$. In 1-based indices (i, j) the reversed matrix has entries $h_{n-1-i+j}$ (first case), which is the Jacobi–Trudi matrix $(h_{\lambda_i - i + j})$ for the constant partition $\lambda_i = n - 1$; by the Jacobi–Trudi identity ([19, I.(3.4)]) its determinant is $s_{((n-1)^n)}$. The shifted case is identical with $\lambda_i = n$. $\square$

Corollary 1 (Normality of the Padé table of E) *Let $\text{ev}_c : \mathfrak{B}_m \rightarrow \mathbb{C}$ be evaluation at $c \in \mathbb{C}^m$ and let $\mathcal{L}$ be the moment functional $\mathcal{L}[x^N] = \text{ev}_c(\Lambda_N)$. The Hankel determinants governing the diagonal Padé approximants of $E(v)$ at c (Gragg [13]) are the numbers $(-1)^{\binom{n}{2}} s_{((n-1)^n)}$ and $(-1)^{\binom{n}{2}} s_{(n^n)}$ evaluated under $p_j = jc_j$ ($j \leq m$), $p_j = 0$ ($j > m$). In particular, for $m = 1$ (so $p_1 = c_1$, $p_{\geq 2} = 0$), the exponential specialization $s_\lambda \mapsto f^\lambda c_1^{|\lambda|} / |\lambda|!$ ([19, I.§7, Ex. 24(a) with $n \rightarrow \infty$; equivalently ev of [19, I.(5.16)] at the exponential specialization]), where $f^\lambda > 0$ is the number of standard Young tableaux, shows every such determinant is nonzero for $c_1 \neq 0$: the Padé table of $e^{c_1 v}$ is normal, recovering the classical normality of the exponential.*

Proof Immediate from Theorem 8, the Padé–Hankel dictionary [13], and $f^\lambda \geq 1$ for every partition λ . $\square$

Remark 4 For $m \geq 2$ the quasi-definiteness locus of $\mathcal{L}$ is precisely the complement of the union of the hypersurfaces $\{c \in \mathbb{C}^m : s_{((n-1)^n)}(p_j = jc_j) = 0\}$; the obstruction to Padé normality of exponentials of principal parts is a rectangular Schur divisor. This gives the “discriminant locus” appearing in the companion Chebyshev–Wheeler paper a name in the pure-pole case.

4 Automorphisms and differential Galois structure

Theorem 9 (Graded Hopf automorphisms) $\text{Aut}_{\text{gr-Hopf}}(\mathfrak{B}_m) \cong (\mathbb{C}^\times)^m$, acting by $c_j \mapsto \lambda_j c_j$. The subgroup induced by the geometric rescalings $w \mapsto \lambda w$ of the local coordinate is the one-parameter diagonal $\lambda_j = \lambda^j$.

Proof A graded Hopf automorphism preserves $\text{Prim}(\mathfrak{B}_m)$ and each of its graded pieces, which are the lines $\mathbb{C}c_j$ by Theorem 3; hence $c_j \mapsto \lambda_j c_j$ with $\lambda_j \in \mathbb{C}^\times$, and any such assignment plainly defines a graded Hopf automorphism. Under $w \mapsto \lambda w$ the principal part $\sum c_j w^{-j}$ becomes $\sum (c_j \lambda^{-j}) w^{-j}$, giving the stated one-parameter subgroup (with λ^{-j} ; reparametrize). $\square$

Proposition 2 (Picard–Vessiot group) Let $R \in \mathbb{C}(z)$ be nonconstant. Then $y = e^R$ is transcendental over $\mathbb{C}(z)$ and the Picard–Vessiot group of $y' = R'y$ over $\mathbb{C}(z)$ is $\mathbb{G}_m$. The group acts on solutions by $y \mapsto \lambda y$, hence on the full Laurent coefficient sequence (A_n) of Section 5 diagonally by $A_n \mapsto \lambda A_n$: the coefficient sequence spans the weight-one isotypic line, while the algebra $\mathfrak{B}_m$ itself is Galois-invariant.

Proof The PV group of a rank-one equation is a Zariski-closed subgroup of $\mathbb{G}_m$, hence $\mathbb{G}_m$ itself or the finite group μ_n ; the latter occurs iff $y^n \in \mathbb{C}(z)$ for some $n \geq 1$ (*van der Put–Singer* [26, Ex. 1.19, §1.4]). Suppose $e^{nR} = f \in \mathbb{C}(z)$. Then $nR' = f'/f$. Every residue of R' vanishes, since R' is the derivative of a rational function; every residue of f'/f at a zero or pole of f is the (nonzero) integer order of f there. Hence f has no zeros or poles, so f is constant and $R' = 0$, contradicting R nonconstant. $\square$

5 The analytic model

We now verify that the formal object of Sections 1–3 computes actual Laurent coefficients, with quantitative control. Fix $c = (c_1, \dots, c_m) \in \mathbb{C}^m$ and write $\Lambda_N(c) := \text{ev}_c(\Lambda_N)$, $C := \sum_{j=1}^m |c_j|$.

Proposition 3 (Growth bound) For all $N \geq 0$ and all $v > 0$, $|\Lambda_N(c)| \leq \Lambda_N(|c_1|, \dots, |c_m|) \leq v^{-N} \exp\left(\sum_j |c_j| v^j\right)$. Consequently, for $N \geq mC$,

$$|\Lambda_N(c)| \leq e^{N/m} \left(\frac{mC}{N}\right)^{N/m},$$

so $|\Lambda_N(c)|^{1/N} \rightarrow 0$ super-geometrically, at the rate of an entire function of order m in $1/w$.

Proof The first inequality is termwise from (2). For the second, all terms of $\sum_M \Lambda_M(|c|) v^M = \exp(\sum_j |c_j| v^j)$ are nonnegative, so any single term is bounded by the sum: $\Lambda_N(|c|) v^N \leq \exp(\sum_j |c_j| v^j) \leq e^{C v^m}$ for $v \geq 1$. Choose $v = (N/(mC))^{1/m} \geq 1$; then $e^{C v^m} = e^{N/m}$ and $v^{-N} = (mC/N)^{N/m}$. $\square$

Theorem 10 (Pure-pole expansion) For every $c \in \mathbb{C}^m$ and $w \neq 0$, the series $\sum_{N \geq 0} \Lambda_N(c) w^{-N}$ converges absolutely, uniformly on $|w| \geq \epsilon$ for each $\epsilon > 0$, and

$$\exp\left(\sum_{j=1}^m c_j w^{-j}\right) = \sum_{N \geq 0} \Lambda_N(c) w^{-N}.$$

In particular $\Lambda_N(c)$ is the coefficient of w^{-N} in the Laurent expansion of e^Π on $\mathbb{C}^\times$, and the map $c \mapsto (\Lambda_N(c))_{N \geq 0}$ is injective.

Proof Expanding each factor $\exp(c_j w^{-j})$ and multiplying, the resulting multi-series is dominated termwise by $\prod_j \sum_{k_j} |c_j|^{k_j} |w|^{-j k_j} / k_j! = \exp(\sum_j |c_j| |w|^{-j}) < \infty$, so it converges absolutely and may be rearranged by total degree, yielding $\sum_N \Lambda_N(c) w^{-N}$ with the coefficients of Lemma 2. Uniqueness of Laurent coefficients on an annulus identifies them with the Laurent coefficients of e^Π . Injectivity: by the triangularity in the proof of Theorem 1, c_N is a polynomial in $\Lambda_1(c), \dots, \Lambda_N(c)$. $\square$

Theorem 11 (Local Laurent structure of e^R ; the coefficient module) *Let $R \in \mathbb{C}(z)$ have a pole at a of order $m \geq 1$ with principal part $\Pi(w) = \sum_{j=1}^m c_j w^{-j}$, $w = z - a$, $c_m \neq 0$, and let $r_a \in (0, \infty]$ be the distance from a to the nearest other pole of R . Write $H := R - \Pi$, analytic on $|w| < r_a$, and $e^{H(a+w)} = \sum_{l \geq 0} b_l w^l$, $\limsup_l |b_l|^{1/l} \leq 1/r_a$. Then on $0 < |w| < r_a$,*

$$e^{R(a+w)} = \sum_{n \in \mathbb{Z}} A_n w^n, \quad A_n = \sum_{k \geq \max(0, -n)} \Lambda_k(c) b_{n+k},$$

each A_n given by an absolutely convergent series. The map $\mathfrak{B}_m \times \mathbb{C}[[w]]_{r_a} \rightarrow \mathbb{C}^\mathbb{Z}$ so defined is $\mathbb{C}$ -bilinear in the coefficient sequences $(\Lambda_k(c))$ and (b_l) : the Laurent coefficients of e^R constitute a module over the coefficient algebra, with structure constants supported on the exponent lattice $n = l - k$.

Proof $e^R = e^\Pi e^H$ exactly, since Π and H are scalar functions. On any circle $|w| = \rho \in (0, r_a)$, both factors are given by absolutely convergent Laurent/Taylor series (Theorem 10; Cauchy), so their product's Laurent coefficients are the Cauchy convolutions $A_n = \sum_k \Lambda_k(c) b_{n+k}$ (with $b_{<0} := 0$), and the double series converges absolutely because, choosing $\rho' \in (\rho, r_a)$, $|b_{n+k}| \leq B \rho'^{-(n+k)}$ while $|\Lambda_k(c)|$ decays super-geometrically by Proposition 3: $\sum_k |\Lambda_k(c)| B \rho'^{-(n+k)} \leq B \rho'^{-n} \sum_k e^{k/m} (mC/k)^{k/m} \rho'^{-k} < \infty$. Bilinearity is manifest. $\square$

Proposition 4 (Truncation error) *In the setting of Theorem 11, fix $\rho' < r_a$ with $|b_l| \leq B \rho'^{-l}$, and let $K \geq K_0 := \max\{mC, e mC (2/\rho')^m\}$. Then*

$$\left| A_n - \sum_{k=\max(0, -n)}^K \Lambda_k(c) b_{n+k} \right| \leq B \rho'^{-n} 2^{-K}.$$

Proof For $k \geq K_0$, Proposition 3 gives $|\Lambda_k(c)| \rho'^{-k} \leq [(e mC/k)^{1/m} / \rho']^k \leq 2^{-k}$ by the choice of K_0 . Sum the geometric tail $\sum_{k > K} 2^{-k} \leq 2^{-K}$ against $B \rho'^{-n}$ (absorbing $\rho'^{-k} \leq$ what was already used). $\square$

Proposition 5 (Hadamard closure) *For $c, c' \in \mathbb{C}^m$, the Hadamard product sequence $(\Lambda_N(c) \Lambda_N(c'))_N$ is P -recursive (holonomic).*

Proof Each factor satisfies the polynomial-coefficient recurrence (1); the Hadamard product of P -recursive sequences is P -recursive (Stanley [25]; Lipshitz [18]). $\square$

6 Classical specializations

The identifications below are exact and exhaustive in the stated cases; no claim is made that a general principal part of order $m \geq 3$ reduces to a single classical one-variable function. The general case is the algebra itself.

Theorem 12 ($m = 1$: modified Bessel) *For $b, c \in \mathbb{C}$, the Laurent expansion of $e^{c/w+bw}$ on $\mathbb{C}^\times$ has coefficients*

$$[w^{-n}] e^{c/w+bw} = \sum_{j \geq 0} \frac{c^{j+n} b^j}{j! (j+n)!} \quad (n \geq 0), \quad [w^n] e^{c/w+bw} = \text{same with } b \leftrightarrow c.$$

For any determination $\sqrt{bc}$ of a square root of bc with $(\sqrt{bc})^2 = bc$, these equal $(c/b)^{n/2} I_n(2\sqrt{bc})$ (interpreting $(c/b)^{n/2} := c^n (bc)^{-n/2}$ with the same determination), where $I_n(x) = \sum_{j \geq 0} (x/2)^{2j+n} / (j! (j+n)!)$. Setting $b = c = x/2$, $w = 1/t$ recovers the classical generating identity $\exp(\frac{x}{2}(t+1/t)) = \sum_{n \in \mathbb{Z}} I_n(x) t^n$.

Proof Here $\Lambda_k(c) = c^k/k!$ and $b_l = b^l/l!$; Theorem 11 with $r_a = \infty$ gives $A_{-n} = \sum_{k \geq n} \frac{c^k}{k!} \frac{b^{k-n}}{(k-n)!}$; substitute $k = j + n$. The Bessel form is the same series regrouped: $\sum_j (bc)^j / (j! (j+n)!)$ = $(bc)^{-n/2} I_n(2\sqrt{bc})$ term by term, both sides being unambiguous power series in bc once the prefactor convention is fixed. Symmetry $w \mapsto 1/w$ exchanges b, c and $n \mapsto -n$, giving the positive coefficients and, with $I_{-n} = I_n$, the classical identity (cf. [7, 10.35.1]). $\square$

Theorem 13 ($m = 2$: Hermite-type polynomials) *Define $\mathbf{H}_N(x, y)$ by $\exp(xt + yt^2) = \sum_{N \geq 0} \mathbf{H}_N(x, y) t^N / N!$ (the Hermite–Kampé de Fériet polynomials [1]). Then for $m = 2$,*

$$\Lambda_N(c_1, c_2) = \frac{\mathbf{H}_N(c_1, c_2)}{N!} = \sum_{0 \leq k \leq N/2} \frac{c_1^{N-2k} c_2^k}{(N-2k)! k!},$$

and $\mathbf{H}_N(x, y) = (i\sqrt{y})^N H_N(x/(2i\sqrt{y}))$ for any fixed square root, where H_N is the classical (physicists') Hermite polynomial.

Proof The first two equalities are Definition 1 for $m = 2$ and Lemma 2. The classical relation follows by comparing $\exp(xt + yt^2)$ with $\exp(2Xs - s^2) = \sum H_N(X) s^N / N!$ under $s = i\sqrt{y}t$, $X = x/(2i\sqrt{y})$; both sides are polynomial identities in $(x, y^{1/2})$, hence hold for every branch choice consistently. $\square$

Theorem 14 (Single-term pole of order m : Wright functions) *Let $\phi(\rho, \beta; \zeta) := \sum_{k \geq 0} \frac{\zeta^k}{k! \Gamma(\rho k + \beta)}$ be Wright's generalized Bessel function (Wright [27]; Kilbas–Saigo [15, §1.11]), entire in ζ for $\rho > -1$. Then for $b \in \mathbb{C}^\times$, $c \in \mathbb{C}$, $n \in \mathbb{Z}$,*

$$[w^{-n}] \exp\left(\frac{c}{w^m} + bw\right) = b^{-n} \phi(m, 1-n; cb^m).$$

Proof Here $\Lambda_{mk} = c^k/k!$ and $\Lambda_N = 0$ otherwise, $b_l = b^l/l!$ ($l \geq 0$), $b_l := 0$ ($l < 0$). Theorem 11: $A_{-n} = \sum_k \frac{c^k}{k!} \frac{b^{mk-n}}{(mk-n)!}$ over k with $mk \geq n$. Since $1/\Gamma(\ell) = 0$ for $\ell \in \mathbb{Z}_{\leq 0}$, the constraint is absorbed by writing $(mk-n)! = \Gamma(mk-n+1) = \Gamma(mk+(1-n))$, and the full sum over $k \geq 0$ equals $b^{-n} \sum_k (cb^m)^k / (k! \Gamma(mk+1-n))$. $\square$

Remark 5 (Scope of special-function reductions) Theorems 12–14 exhaust the cases in which a single classical family arises from a natural one-parameter entire factor: $m = 1$, $m = 2$, and pure top-order poles. For a general principal part with $m \geq 3$ and mixed c_j , the coefficient A_n is the genuine multivariate partition sum of Theorem 11, i.e. an element of the $\mathfrak{B}_m$ -module, and we do not assert any reduction to Wright, Fox–Wright, or Fox H functions of one variable. What replaces such a reduction is the structure theory of Sections 2–3 together with the holonomy of Section 7: closed recurrences, not closed one-variable forms.

7 Holonomy

Theorem 15 (D-finiteness and the coefficient recurrence) *Let $R \in \mathbb{C}(z)$ be nonconstant, write $R' = U/V$ in lowest terms with $U, V \in \mathbb{C}[z]$, and let a be a pole of R of order $m \geq 1$. Set $\tilde{V}(w) := V(a+w) = \sum_{p=0}^{d_V} V_p w^p$, $\tilde{U}(w) := U(a+w) = \sum_{p=0}^{d_U} U_p w^p$. Then $y = e^R$ satisfies the first-order equation $Vy' - Uy = 0$ over $\mathbb{C}(z)$ (in particular y is D-finite), and the full two-sided Laurent coefficient sequence $(A_n)_{n \in \mathbb{Z}}$ of y at a (Theorem 11) satisfies, for every $s \in \mathbb{Z}$,*

$$\sum_{p=0}^{d_V} V_p (s+1-p) A_{s+1-p} = \sum_{p=0}^{d_U} U_p A_{s-p}. \quad (3)$$

This is a linear recurrence with coefficients affine in s , of order at most $\max(d_V, d_U + 1)$, valid in both directions $n \rightarrow \pm\infty$. Moreover $w^{m+1} \mid \tilde{V}$ (indeed $\text{ord}_{w=0} R' = -(m+1)$ forces $\text{ord } \tilde{V} - \text{ord } \tilde{U} = m+1$), and in the pure-pole case $R = \Pi$ the recurrence (3) reduces exactly to the universal recurrence (1).

Proof $y' = R'y$ is immediate; clearing denominators gives $Vy' = Uy$ with polynomial coefficients, which is the definition of D-finiteness for y (Stanley [25]). On $0 < |w| < r_a$ insert $y = \sum_n A_n w^n$ (absolutely and locally uniformly convergent, so termwise differentiation and Cauchy products with the polynomials $\tilde{V}, \tilde{U}$ are valid): $\sum_p V_p w^p \sum_n n A_n w^{n-1} = \sum_p U_p w^p \sum_n A_n w^n$. Equate coefficients of w^s : $\sum_p V_p (s+1-p) A_{s+1-p} = \sum_p U_p A_{s-p}$, which is (3). The order statement counts the index span. The valuation claim: near a , $R' = -mc_m w^{-(m+1)} + O(w^{-m})$ with $c_m \neq 0$, so $\text{ord}_{w=0}(\tilde{U}/\tilde{V}) = -(m+1)$; since $\gcd(U, V) = 1$, at most one of $\tilde{U}(0), \tilde{V}(0)$ vanishes, forcing $\tilde{U}(0) \neq 0$ and $w^{m+1} \parallel \tilde{V}$. Pure pole: $R = \Pi$ gives $R' = -\sum_j j c_j w^{-j-1} = \tilde{U}/\tilde{V}$ with $\tilde{V} = w^{m+1}$, $\tilde{U} = -\sum_{j=1}^m j c_j w^{m-j}$; then (3) reads $(s-m)A_{s-m} = -\sum_j j c_j A_{s-m+j}$, and putting $A_{-N} = \Lambda_N$, $s-m = -N$: $N\Lambda_N = \sum_j j c_j \Lambda_{N-j}$. $\square$

Corollary 2 *For fixed c , every coefficient sequence in the module of Theorem 11 is P-recursive with an explicitly computable recurrence (3); creative-telescoping toolchains apply directly (Kauers–Paule [14]; Koutschan [16]).*

8 Axiomatic characterization

Following the constructive development, we now characterize what has been built. Two statements are proved: an *initiality* theorem in the category naturally attached to the universal recurrence, and a *recognition* theorem identifying $\mathfrak{B}_m$ among graded Hopf algebras. The hypotheses of the recognition theorem are shown to be independent.

8.1 Initiality

Definition 3 (The category $\mathcal{C}_m$) *Objects of $\mathcal{C}_m$ are triples $(A, \gamma = (\gamma_1, \dots, \gamma_m) \in A^m, (\lambda_N)_{N \geq 0})$ where A is a commutative unital $\mathbb{C}$ -algebra, $\lambda_0 = 1$, and*

$$N\lambda_N = \sum_{j=1}^{\min(m, N)} j \gamma_j \lambda_{N-j} \quad (N \geq 1). \quad (4)$$

A morphism $(A, \gamma, \lambda) \rightarrow (A', \gamma', \lambda')$ is a unital $\mathbb{C}$ -algebra homomorphism $\varphi : A \rightarrow A'$ with $\varphi(\gamma_j) = \gamma'_j$ for all j and $\varphi(\lambda_N) = \lambda'_N$ for all N .

Theorem 16 (Initiality) *$(\mathfrak{B}_m, c, \Lambda)$ is an initial object of $\mathcal{C}_m$: for every object (A, γ, λ) there is exactly one morphism $\varphi : (\mathfrak{B}_m, c, \Lambda) \rightarrow (A, \gamma, \lambda)$.*

Proof *Existence.* Since $\mathfrak{B}_m$ is the free commutative $\mathbb{C}$ -algebra on $c_1, \dots, c_m$, there is a unique algebra map φ with $\varphi(c_j) = \gamma_j$. We claim $\varphi(\Lambda_N) = \lambda_N$ for all N , by strong induction: true at $N = 0$ ($\Lambda_0 = 1 \mapsto 1 = \lambda_0$); for $N \geq 1$, since N is invertible in the $\mathbb{C}$ -algebra A ,

$$\varphi(\Lambda_N) = \varphi \left(\frac{1}{N} \sum_j j c_j \Lambda_{N-j} \right) = \frac{1}{N} \sum_j j \gamma_j \varphi(\Lambda_{N-j}) = \frac{1}{N} \sum_j j \gamma_j \lambda_{N-j} = \lambda_N,$$

using Lemma 1 in $\mathfrak{B}_m$, the inductive hypothesis, and (4) in A . *Uniqueness.* Any morphism must send $c_j \mapsto \gamma_j$, and an algebra map out of a polynomial ring is determined by the images of the variables; the λ -compatibility is then automatic by the computation just made, so no further freedom exists. $\square$

Remark 6 Equation (4) determines (λ_N) uniquely from γ in any $\mathbb{C}$ -algebra; so $\mathcal{C}_m$ is equivalent to the category of m -pointed commutative $\mathbb{C}$ -algebras, and initiality is exactly the freeness of $\mathfrak{B}_m$ together with the verification that the distinguished sequence transports along the unique map. The content is modest but the statement is now a theorem with a defined category, defined morphisms, and a proof, rather than an assertion.

8.2 Recognition

Theorem 17 (Recognition) *Let A be a graded connected commutative $\mathbb{C}$ -Hopf algebra with a sequence $\lambda_N \in A_N$, $\lambda_0 = 1$, satisfying:*

- (G) *the λ_N generate A as an algebra;*
- (D) *$\Delta \lambda_N = \sum_{i=0}^N \lambda_i \otimes \lambda_{N-i}$ for all N (divided-power sequence);*

(T) $\text{Prim}(A)_j = 0$ for all $j > m$;

(I) $\lambda_1, \dots, \lambda_m$ are algebraically independent over $\mathbb{C}$.

Then there is a unique isomorphism of graded Hopf algebras $\varphi : \mathfrak{B}_m \rightarrow A$ with $\varphi(\Lambda_N) = \lambda_N$ for all N ; and $\varphi(c_j)$ is the weight- j coefficient of $\log \sum_N \lambda_N t^N$.

Proof Set $\Lambda^A(t) := \sum_{N \geq 0} \lambda_N t^N \in A[[t]]$. By (D), $\Lambda^A(t)$ is group-like in the completed Hopf algebra $A[[t]]$ (coproduct and counit extended t -adically): $\Delta \Lambda^A = \Lambda^A \otimes \Lambda^A$ read in $(A \otimes A)[[t]]$, $\varepsilon \Lambda^A = 1$. Since $\Lambda^A(t) - 1 \in tA[[t]]$, the logarithm $\ell(t) := \log \Lambda^A(t) = \sum_{j \geq 1} \gamma_j t^j$ is defined, and

$$\Delta \ell = \log((\Lambda^A \otimes 1)(1 \otimes \Lambda^A)) = \log(\Lambda^A \otimes 1) + \log(1 \otimes \Lambda^A) = \ell \otimes 1 + 1 \otimes \ell,$$

the middle equality because the two arguments commute. Hence every γ_j is primitive; also $\gamma_j \in A_j$ because $\lambda_N \in A_N$ and $\log$ preserves the grading of coefficients. By (T), $\gamma_j = 0$ for $j > m$, so

$$\Lambda^A(t) = \exp \left(\sum_{j=1}^m \gamma_j t^j \right), \quad \text{i.e.} \quad \lambda_N = \Lambda_N(\gamma_1, \dots, \gamma_m).$$

Define the algebra map $\varphi : \mathfrak{B}_m \rightarrow A$ by $\varphi(c_j) = \gamma_j$; then $\varphi(\Lambda_N) = \lambda_N$ by the displayed identity (or by Theorem 16, since $(A, \gamma, \lambda) \in \mathcal{C}_m$: apply d/dt to $\Lambda^A = \exp(\sum \gamma_j t^j)$ as in Lemma 1). φ is a morphism of graded Hopf algebras: it preserves grading ($\gamma_j \in A_j$) and sends the primitive generators c_j to primitives, which determines compatibility with Δ, ε, S on all of $\mathfrak{B}_m$ since these are (anti)algebra maps. *Surjectivity*: the image contains every λ_N , which generate by (G). *Injectivity*: $\lambda_1, \dots, \lambda_m$ are polynomials in $\gamma_1, \dots, \gamma_m$ and conversely (the triangularity of Theorem 1 transports along φ), so (I) is equivalent to algebraic independence of $\gamma_1, \dots, \gamma_m$; an algebra map from $\mathbb{C}[c_1, \dots, c_m]$ sending the variables to algebraically independent elements has zero kernel. *Uniqueness*: any graded Hopf map ψ with $\psi(\Lambda_N) = \lambda_N$ satisfies $\psi(\log E(t)) = \ell(t)$ coefficient-wise, i.e. $\psi(c_j) = \gamma_j$, and is therefore φ . $\square$

8.3 Independence of the hypotheses

Proposition 6 *Each of (G), (D), (T), (I) is independent of the remaining three: for each hypothesis there is a triple $(A, (\lambda_N))$ satisfying the other three but not isomorphic to $(\mathfrak{B}_m, (\Lambda_N))$ in the sense of Theorem 17.*

Proof Fix $m \geq 2$ (for $m = 1$ replace the (I)-example by $A = \mathbb{C}$, $\lambda_N = \delta_{N,0}$, which satisfies (G),(D),(T) but not (I)).

Drop (G). $A = \mathfrak{B}_m[y]$, y primitive of weight 1, with $\lambda_N := \Lambda_N$. (D) holds as before; $\text{Prim}(A)_j = \mathbb{C}c_j$ for $2 \leq j \leq m$, $\text{Prim}(A)_1 = \mathbb{C}c_1 \oplus \mathbb{C}y$, and 0 for $j > m$, so (T) holds; (I) holds. But the λ_N do not generate (y is missing), and $A \not\cong \mathfrak{B}_m$ carrying the sequences (Hilbert series differ).

Drop (I). $A = \mathfrak{B}_m/(c_m) \cong \mathfrak{B}_{m-1}$ with $\lambda_N := \text{image of } \Lambda_N$, i.e. $\Lambda_N(c_1, \dots, c_{m-1}, 0)$. (c_m) is a Hopf ideal (generated by a primitive), so A is a graded connected Hopf algebra and (D) descends; (G) descends; (T) holds a fortiori since $\text{Prim}(A)_j = 0$ for $j \geq m$. But $\lambda_1, \dots, \lambda_m$ satisfy the algebraic relation expressing λ_m through $\lambda_1, \dots, \lambda_{m-1}$ (set $c_m = 0$ in $\Lambda_m = c_m + q_m(c_1, \dots, c_{m-1})$), and $A \not\cong \mathfrak{B}_m$ (transcendence degree $m - 1$ vs. m).

Drop (T). $A = \mathfrak{B}_{m+1}$ with its own sequence $\lambda_N := \Lambda_N^{(m+1)}$. (G), (D), (I) hold (Theorems 1, 2; independence of the first m among the first $m+1$). $\text{Prim}(A)_{m+1} = \mathbb{C}c_{m+1} \neq 0$, and no isomorphism onto $\mathfrak{B}_m$ exists (transcendence degree).

Drop (D). $A = \mathfrak{B}_m$ as a graded Hopf algebra (so (T) is unchanged), with the perturbed sequence $\lambda'_N := \Lambda_N$ for $N \neq 2$ and $\lambda'_2 := \Lambda_2 + c_1^2$. (G): $\lambda'_1 = c_1$ and $\lambda'_2 - \frac{3}{2}\lambda_1'^2 = c_2$ recover the generators, and inductively all c_j ; (I): $\lambda'_1, \dots, \lambda'_m$ differ from $\Lambda_1, \dots, \Lambda_m$ by a weight-preserving triangular polynomial change, preserving independence. But

$$\Delta\lambda'_2 - \sum_{i=0}^2 \lambda'_i \otimes \lambda'_{2-i} = \Delta c_1^2 - (c_1^2 \otimes 1 + 1 \otimes c_1^2) = 2c_1 \otimes c_1 \neq 0,$$

so (D) fails; and no isomorphism φ with $\varphi(\Lambda_N) = \lambda'_N$ can exist, since a Hopf map would transport identity (D), which holds for (Λ_N) and fails for (λ'_N) . $\square$

Remark 7 The axioms now do work in the promised sense: (D)+(T) alone force the generating series into the exponential normal form $\exp(\sum_{j \leq m} \gamma_j t^j)$ (proof of Theorem 17), i.e. they *derive* the universal recurrence rather than postulate it; (G)+(I) then pin the algebra. Grading, derivations (Lemma 3, recovered as ∂_{γ_j}), and the product/coproduct formulas are theorems from the axioms, and the Laurent construction of Section 5 is a model by Theorem 10.

9 Application: rough Heston pricing in the coefficient algebra

The rough Heston log-price characteristic function (*El Euch–Rosenbaum [8]*) is $\varphi_T(a) = \exp(L(a, T))$, where $X_T := \log(S_T/S_0) - rT$ is the driftless discounted log-price, a is a complex moment/Fourier variable, and

$$L(a, t) := \theta \int_0^t h(s, a) \, ds + V_0 I^{1-\alpha} h(t, a), \quad D^\alpha h = P(a) + Q(a)h + R(a)h^2, \quad I^{1-\alpha} h(0, a) = 0,$$

$\alpha = H + \frac{1}{2} \in (0, 1)$, $(P, Q, R)(a) = (\frac{1}{2}(-a^2 - ia), ia\rho\nu - \lambda, \frac{1}{2}\nu^2)$, and I^ρ denotes Riemann–Liouville fractional integration.

9.1 The fractional Riccati equation: exact solution and diagonal Padé approximants

Lemma 4 (Müntz–Tau expansion) *The fractional IVP $D^\alpha h = P(a) + Q(a)h + R(a)h^2$, $I^{1-\alpha} h(0, a) = 0$, is equivalent to the Volterra equation $h = I^\alpha[P(a) + Q(a)h + R(a)h^2]$, has a unique solution on $[0, \infty)$ for every a , and that solution is $h(t, a) = \sum_{k \geq 1} c_k(a) t^{k\alpha}$ with*

$$c_1(a) = \frac{P(a)}{\Gamma(\alpha + 1)}, \quad c_{k+1}(a) = \frac{\Gamma(k\alpha + 1)}{\Gamma((k+1)\alpha + 1)} \left(Q(a)c_k(a) + R(a) \sum_{\substack{i+j=k \\ i, j \geq 1}} c_i(a)c_j(a) \right), \quad k \geq 1.$$

Proof Applying I^α to $D^\alpha h = P + Qh + Rh^2$ and using $I^\alpha D^\alpha h = h$ (valid under $I^{1-\alpha}h(0) = 0$, the Riemann–Liouville composition rule) gives the Volterra form. On $[0, T]$, the map $\Phi(h) := I^\alpha[P + Qh + Rh^2]$ is Lipschitz on bounded sets (the quadratic term is locally Lipschitz), and in the weighted norm $\|h\|_\beta := \sup_{t \in [0, T]} |e^{-\beta t} h(t)|$ the weakly singular kernel of I^α contributes a factor that $\rightarrow 0$ as $\beta \rightarrow \infty$ (the standard fractional Grönwall estimate: $|I^\alpha f(t)| \leq \frac{1}{\Gamma(\alpha)} \int_0^t (t-s)^{\alpha-1} |f(s)| ds \leq \|f\|_\beta \frac{1}{\Gamma(\alpha)} \int_0^t (t-s)^{\alpha-1} e^{\beta s} ds$, and the last integral, divided by $e^{\beta t}$, is $O(\beta^{-\alpha})$), so Φ is a contraction for β large enough, giving a unique fixed point on $[0, T]$ by Banach’s theorem; letting $T \rightarrow \infty$ (the fixed points on nested intervals agree by uniqueness) gives global existence and uniqueness on $[0, \infty)$. Substituting the ansatz $h = \sum_{k \geq 1} c_k t^{k\alpha}$ into the Volterra form and using $I^\alpha t^s = \frac{\Gamma(s+1)}{\Gamma(s+\alpha+1)} t^{s+\alpha}$ termwise: $I^\alpha P(a) = \frac{P(a)}{\Gamma(\alpha+1)} t^\alpha$ matches the coefficient of t^α , giving c_1 ; $I^\alpha[Qh + Rh^2] = \sum_{k \geq 1} \frac{\Gamma(k\alpha+1)}{\Gamma((k+1)\alpha+1)} \left(Qc_k + R \sum_{i+j=k} c_i c_j \right) t^{(k+1)\alpha}$, and matching the coefficient of $t^{(k+1)\alpha}$ gives c_{k+1} ; by uniqueness of the fixed point, this power series (which converges on $[0, \infty)$ by Lemma 10 below) is exactly h . $\square$

Lemma 5 (Polynomiality) $c_k(a) \in \mathbb{C}[a]$ for every $k \geq 1$.

Proof Induction on k : $c_1 = P(a)/\Gamma(\alpha+1)$ is a polynomial since P is. If $c_1, \dots, c_k$ are polynomials, then c_{k+1} , a scalar multiple of $Q(a)c_k(a) + R(a) \sum_{i+j=k} c_i(a)c_j(a)$, is a $\mathbb{C}$ -linear combination of products of polynomials in a , hence a polynomial. $\square$

Definition 4 (Moment functional) For fixed a , $m(k, a) := c_{k+1}(a)$, $k \geq 0$; $\mathcal{L}_a : \mathbb{C}[X] \rightarrow \mathbb{C}[a]$, $\mathcal{L}_a[X^k] := m(k, a)$, extended $\mathbb{C}$ -linearly. $H_n(a) := \det[m(i+j, a)]_{0 \leq i, j \leq n-1} \in \mathbb{C}[a]$.

Lemma 6 (Signed quasi-definiteness) $H_n(a) \neq 0$ as a polynomial in a for every $n \geq 1$ (so $\bigcup_n \{a : H_n(a) = 0\}$ is a countable union of finite sets). For a real with $\rho\nu \neq 0$, $\mathcal{L}_a$ is not positive-definite on $\mathbb{R}[X]$: there is $p \in \mathbb{R}[X]$ with $\mathcal{L}_a[p^2] \notin \mathbb{R}_{\geq 0}$.

Proof $m(0, a) = c_1(a) = P(a)/\Gamma(\alpha+1) \neq 0$ gives $H_1 \neq 0$; for $n \geq 2$, the leading a -degree term of H_n , formed from the highest-degree contributions of each $m(k, a)$ under the explicit recurrence of Lemma 4, is a nonzero monomial (a Hankel determinant of a sequence with strictly increasing polynomial degree has leading term the product of the individual leading terms along the anti-diagonal, since lower-degree cross terms cannot cancel a top-degree contribution appearing in only one placement of the determinant expansion), so $H_n \neq 0$; its zero set is a proper algebraic subvariety of $\mathbb{C}$, i.e. finite. For $\mathcal{L}_a$ with $a \in \mathbb{R} \setminus \{0\}$ and $\rho\nu \neq 0$: $\Im Q(a) = a\rho\nu \neq 0$, so $Q(a) \notin \mathbb{R}$, while $R = \frac{1}{2}\nu^2 \in \mathbb{R}_{\geq 0}$ and $P(a) \in \mathbb{R}$ for real a (since $-a^2 - ia$ has real part $-a^2$ for real a ... more precisely $P(a) = \frac{1}{2}(-a^2 - ia)$ has nonzero imaginary part $-a/2$ for $a \neq 0$, so $P(a) \notin \mathbb{R}$ either). The full moment sequence $m(k, a)$ is then genuinely complex-valued for every $k \geq 0$ with a nonzero imaginary part not forced to cancel (each $c_k(a)$, being a nontrivial polynomial combination of $P(a), Q(a), R$ with positive integer coefficients arising from the recurrence, inherits a nonzero imaginary part generically). A real, symmetric, positive-definite bilinear form on $\mathbb{R}[X]$ has real Hankel determinants of a fixed sign pattern ($H_n > 0$ for all n); since $m(k, a) \notin \mathbb{R}$ for $a \neq 0$, $H_n(a) \notin \mathbb{R}$ for generic such n , which is incompatible with positive-definiteness on $\mathbb{R}[X]$ (which forces $\mathcal{L}_a[p^2] \in \mathbb{R}_{\geq 0}$, in particular $\mathcal{L}_a[X^{2j}] \in \mathbb{R}$ for all j — already false for j such that $m(2j, a) \notin \mathbb{R}$). $\square$

Definition 5 (Formal OPS) $\Pi(n, X) \in \mathbb{C}(a)[X]$, monic of degree n , satisfies $\mathcal{L}_a[\Pi(n, X)X^k] = 0$ for $0 \leq k < n$; equivalently, on the complement of $\bigcup_{k \leq n} \{H_k = 0\}$, where Heine's classical determinantal formula

$$\Pi(n, X) = H_n(a)^{-1} \det \begin{pmatrix} m_0 & m_1 & \cdots & m_n \\ m_1 & m_2 & \cdots & m_{n+1} \\ \vdots & & \ddots & \vdots \\ m_{n-1} & m_n & \cdots & m_{2n-1} \\ 1 & X & \cdots & X^n \end{pmatrix}, \quad m_i := m(i, a),$$

produces the unique such monic polynomial, $\Pi(n, X)$ satisfies

$$\Pi(n+1, X) = (X - \alpha(n))\Pi(n, X) - \beta(n)\Pi(n-1, X), \quad \Pi(-1, X) = 0, \quad \Pi(0, X) = 1,$$

for uniquely determined $\alpha(n), \beta(n) \in \mathbb{C}(a)$.

Definition 6 (Generating polynomial) $S(n, x) := \sum_{k \geq 0} \mathcal{L}_a[X^k \Pi(n, X)] x^k \in \mathbb{C}(a)[[x]]$, $S(-1, x) := 1$.

Lemma 7 (Vanishing order and norm) $\text{ord}_x S(n, x) = n$ and $[x^n]S(n, x) = h(n) := \mathcal{L}_a[\Pi(n, X)^2] = H_{n+1}(a)/H_n(a) \in \mathbb{C}(a)^\times$ on the complement of $\bigcup_{k \leq n+1} \{H_k = 0\}$.

Proof Orthogonality gives $\mathcal{L}_a[X^k \Pi(n, X)] = 0$ for $k < n$, so $\text{ord}_x S(n, x) \geq n$. At $k = n$: since $\Pi(n, X)$ is monic, $X^n = \Pi(n, X) + q(X)$ with $\deg q < n$, so $X^n \Pi(n, X) = \Pi(n, X)^2 + q(X)\Pi(n, X)$, and $\mathcal{L}_a[q(X)\Pi(n, X)] = 0$ by orthogonality (expand q in the basis $1, X, \dots, X^{n-1}$); hence $[x^n]S(n, x) = \mathcal{L}_a[\Pi(n, X)^2]$. The identity $h(n) = H_{n+1}/H_n$ follows by substituting Heine's determinantal formula for $\Pi(n, X)$ into $\mathcal{L}_a[\Pi(n, X)^2]$ and expanding the resulting $(n+1) \times (n+1)$ bordered determinant along its last row/column, a standard computation in the theory of formal orthogonal polynomials. $\square$

Theorem 18 (Polynomial Chebyshev–Wheeler recurrence) With $S(-1, x) := 1$, the sequence $\{S(n, x)\}_{n \geq -1}$ obeys, for $n \geq 0$,

$$S(n+1, x) = \frac{S(n, x) - S(n, 0)}{x} - \alpha(n)S(n, x) - \beta(n)S(n-1, x), \quad (5)$$

where $\alpha(n), \beta(n) \in \mathbb{C}(a)$ are the unique scalars for which $\text{ord}_x S(n+1, x) \geq n+1$, explicitly

$$\beta(n) = \frac{h(n)}{h(n-1)}, \quad \alpha(n) = \frac{[x^n](S(n, x)/x) - \beta(n)[x^n]S(n-1, x)}{[x^n]S(n, x)}.$$

Consequently $S(n, x)$ (hence $\alpha(n), \beta(n)$) is computed recursively from $S(0, x) = \sum_{k \geq 0} m(k, a)x^k$ using only polynomial coefficient arithmetic in $\mathbb{C}(a)[x]$ truncated to a working bandwidth: no Hankel determinant H_n is ever formed or inverted.

Proof Apply $\mathcal{L}_a[\cdot X^k]$ to the three-term recurrence $X\Pi(n, X) = \Pi(n+1, X) + \alpha(n)\Pi(n, X) + \beta(n)\Pi(n-1, X)$ and multiply by x^k :

$$\mathcal{L}_a[X^{k+1}\Pi(n, X)] = \mathcal{L}_a[X^k \Pi(n+1, X)] + \alpha(n)\mathcal{L}_a[X^k \Pi(n, X)] + \beta(n)\mathcal{L}_a[X^k \Pi(n-1, X)].$$

Sum over $k \geq 0$ against x^k . The left side telescopes to $\sum_{k \geq 0} \mathcal{L}_a[X^{k+1}\Pi(n, X)]x^k = x^{-1} \sum_{k \geq 1} \mathcal{L}_a[X^k\Pi(n, X)]x^k = x^{-1}(S(n, x) - S(n, 0))$; the right side is $S(n+1, x) + \alpha(n)S(n, x) + \beta(n)S(n-1, x)$. Rearranging gives (5). (For $n \geq 1$, $S(n, 0) = \mathcal{L}_a[\Pi(n, X)] = 0$ by orthogonality against X^0 , so the difference quotient is genuine; for $n = 0$, $S(-1, 0) = 1 \neq 0$ but the recurrence for $n = 0$ only needs $S(-1, x) \equiv 1$, consistent with the displayed formula since the $\beta(0)$ term multiplies $S(-1, \cdot)$, a known constant.) By Lemma 7, $\text{ord}_x S(n, x) = n$ for $n \geq 0$; demanding $\text{ord}_x S(n+1, x) \geq n+1$ in (5) imposes exactly two linear conditions on the two unknowns $\alpha(n), \beta(n)$ — vanishing of the coefficients of x^{n-1} and x^n on the right side — whose unique solution (a 2×2 linear system with the coefficient of $\beta(n)$ in the x^{n-1} equation being $[x^{n-1}]S(n-1, x) = h(n-1) \neq 0$ by Lemma 7) is the displayed formula for $\beta(n)$, after which the x^n equation, linear in $\alpha(n)$ with nonzero coefficient $[x^n]S(n, x) = h(n) \neq 0$, gives $\alpha(n)$. $\square$

Definition 7 (Reciprocal flip) For $p(X) = \sum_{k=0}^n c_k X^k \in \mathbb{C}(a)[X]$, $p^\flat(x) := x^n p(1/x) = \sum_{k=0}^n c_k x^{n-k}$.

Theorem 19 (Padé numerator and denominator) Let $g(x, a) := \sum_{k \geq 1} c_k(a)x^k$, the Müntz–Tau generating series of Lemma 4 written in $x = t^\alpha$. Let $\Pi^{(1)}(n, X)$, the associated (second-kind) polynomial sequence, satisfy the same three-term recurrence as $\Pi(n, X)$ (Definition 5) with seeds $\Pi^{(1)}(-1, X) := -1$, $\Pi^{(1)}(0, X) := 0$. Then the diagonal $[M/M]$ Padé denominator and numerator of g are $\hat{Q}(M, x) = \Pi(M, \cdot)^\flat(x)$ and $\hat{P}(M, x) = \Pi^{(1)}(M, \cdot)^\flat(x)$: explicitly,

$$g(x, a)\hat{Q}(M, x) - x\hat{P}(M, x) = O(x^{2M+2}) \quad \text{in } \mathbb{C}(a)[[x]].$$

Proof Write $g(x, a) = \sum_{j \geq 1} m(j-1, a)x^j$ (Definition 4). Then $g(x, a)x^M \Pi(M, 1/x) = \sum_j m(j-1, a)x^j \cdot x^M \Pi(M, 1/x)$; substituting $\Pi(M, X) = \sum_{i=0}^M \pi_i X^i$ ($\pi_M = 1$) gives $x^M \Pi(M, 1/x) = \sum_i \pi_i x^{M-i}$, so the coefficient of $x^{M+1+\ell}$ ($\ell = 0, \dots, M-1$) in the product is $\sum_i \pi_i m(M-i+\ell, a) = \mathcal{L}_a[\Pi(M, X)X^\ell]$ (reindexing the convolution and recognizing the moment-functional pairing), which vanishes for $0 \leq \ell \leq M-1$ by orthogonality (Definition 5). Hence $g\hat{Q}(M, \cdot)$ agrees with a polynomial of degree $\leq M$ up to order x^{2M+2} ; that polynomial, divided by x , is exactly $\hat{P}(M, x)$: the coefficients of $x^1, \dots, x^M$ in $g\hat{Q}(M, \cdot)$ are $\sum_i \pi_i m(\ell-1-i, a)$ for $\ell = 1, \dots, M$ (with $m(< 0, a) := 0$), which is exactly the divided-difference expression $\mathcal{L}_{a,Y}[(\Pi(M, X) - \Pi(M, Y))/(X - Y)]$ evaluated through its recurrence-defined polynomial coefficients, i.e. $\Pi^{(1)}(M, X)$ flipped; this is a direct computation from the shared three-term recurrence with the stated seeds (the standard associated-polynomial identity for orthogonal polynomial sequences). $\square$

Definition 8 (Jacobi matrix) $J(M, a) \in \mathbb{C}(a)^{M \times M}$ is tridiagonal with diagonal $(\alpha(0), \dots, \alpha(M-1))$, super-diagonal entries 1, and sub-diagonal $(\beta(1), \dots, \beta(M-1))$.

Lemma 8 (Spectrum and pole locus) $\sigma(J(M, a)) = \{X : \Pi(M, X) = 0\}$ as multisets, and the poles of $\hat{P}(M, \cdot)/\hat{Q}(M, \cdot)$ are the reciprocals of $\sigma(J(M, a))$.

Proof In the basis $\{\Pi(0, \cdot), \dots, \Pi(M-1, \cdot)\}$, multiplication by X acts by the companion-matrix realization $X\Pi(n, \cdot) = \Pi(n+1, \cdot) + \alpha(n)\Pi(n, \cdot) + \beta(n)\Pi(n-1, \cdot)$ (Definition 5) for $n < M-1$, and by the same recurrence with the $\Pi(M, \cdot)$ term replaced using $\Pi(M, X) = 0$ at $n = M-1$ once restricted to the quotient by $(\Pi(M, X))$; this is exactly $J(M, a)$, so its characteristic polynomial $\det(XI - J(M, a))$ equals $\Pi(M, X)$. The pole statement is Theorem 19 and Definition 7: zeros of $\hat{Q}(M, x) = x^M \Pi(M, 1/x)$ are the reciprocals of the zeros of $\Pi(M, \cdot)$. $\square$

Theorem 20 (Real-ray pole-freeness) *For every real a , $t \mapsto h(t, a)$ exists and is finite for all $t \in [0, \infty)$ (Lemma 4, global existence). Consequently $g(\cdot, a)$ extends to a function holomorphic on a neighborhood of $[0, \infty)$ in $x = t^\alpha$, and its Stahl compact $\Delta_g(a)$ (the set to which the poles of its diagonal Padé approximants condense) satisfies $\Delta_g(a) \cap [0, \infty) = \emptyset$.*

Proof Global existence on $[0, \infty)$ is Lemma 4. A power series in t^α that converges to a finite value at every $t \in [0, \infty)$, and whose sum solves an algebraic-coefficient ODE (the fractional Riccati equation) locally analytic in t off any singularity, has no singularity on $[0, \infty)$: were there a singular point $t_0 \in (0, \infty)$, the Volterra solution would fail to extend continuously past t_0 , contradicting the global existence already established for every finite window $[0, T]$, $T > t_0$. Since $x = t^\alpha$ is a homeomorphism $[0, \infty) \rightarrow [0, \infty)$, $g(\cdot, a)$ is holomorphic on a neighborhood of $[0, \infty)$ in x ; its Stahl compact, by definition the complement (in $\overline{\mathbb{C}}$) of the largest domain to which g continues with minimal capacity boundary, therefore avoids $[0, \infty)$. $\square$

Theorem 21 (Diagonal Padé convergence) *For every compact $K \subset \mathbb{C} \setminus \Delta_g(a)$, $\hat{P}(M, \cdot)/\hat{Q}(M, \cdot) \rightarrow g(\cdot, a)$ uniformly on K as $M \rightarrow \infty$ (Stahl [24, Thm. 1, §6]), and the pole sets of $\hat{P}(M, \cdot)/\hat{Q}(M, \cdot)$ converge to $\Delta_g(a)$ in capacity. In particular, by Theorem 20, for every $x^* > 0$ there is $M_0 = M_0(a, x^*)$ such that $\hat{Q}(M, \cdot)$ has no zero on $[0, x^*]$ for $M \geq M_0$, and $\hat{P}(M, \cdot)/\hat{Q}(M, \cdot) \rightarrow g(\cdot, a)$ uniformly on $[0, x^*]$.*

Proof Stahl's theorem, applied to the diagonal Padé sequence of Theorem 19. Since $[0, x^*]$ is a fixed compact disjoint from $\Delta_g(a)$ (Theorem 20) and the pole sets converge to $\Delta_g(a)$ in capacity, the standard capacity-to-uniform argument on compacts avoiding the limiting singular set gives: for M large, no pole lies within a fixed positive distance of $[0, x^*]$, and the convergence promised by Stahl's theorem on any compact neighborhood of $[0, x^*]$ avoiding $\Delta_g(a)$ restricts to uniform convergence on $[0, x^*]$ itself. $\square$

Theorem 22 (Integrated and fractionally-integrated series) $Y(t, a) := \int_0^t h(s, a) ds = \sum_{k \geq 1} \frac{c_k(a)}{k\alpha+1} t^{k\alpha+1}$, and for $\rho \geq 0$, $I^\rho h(t, a) = \sum_{k \geq 1} c_k(a) \frac{\Gamma(k\alpha+1)}{\Gamma(k\alpha+1+\rho)} t^{k\alpha+\rho}$. Both are Müntz series in the reweighted moment functional $\mathcal{L}_a^{(\rho)}[X^k] := m(k, a)\Gamma(k\alpha+1)/\Gamma(k\alpha+1+\rho)$, to which Theorems 18–20 apply verbatim (their proofs use only $m(k, a) \in \mathbb{C}(a)$, never the specific values), producing diagonal Padé approximants of Y and of $I^\rho h$ with the same real-ray convergence guarantee.

Proof Termwise integration gives $\int_0^t s^{k\alpha} ds = t^{k\alpha+1}/(k\alpha+1)$. The Riemann–Liouville rule $I^\rho t^{k\alpha} = \frac{\Gamma(k\alpha+1)}{\Gamma(k\alpha+1+\rho)} t^{k\alpha+\rho}$ applied termwise to $h = \sum c_k t^{k\alpha}$ gives the second series (the case $\rho = 1$ recovers the first, since $\Gamma(k\alpha+1)/\Gamma(k\alpha+2) = 1/(k\alpha+1)$). Both series are of the form $\sum_k \tilde{m}(k-1, a)x^k$ for a reweighted moment sequence $\tilde{m}(k, a) = m(k, a) \cdot$ (a scalar depending on k, ρ only); nothing in the proofs of Theorems 18–20 used more about $m(k, a)$ than that it lies in $\mathbb{C}(a)$ and the resulting Hankel determinants are generically nonzero (true for $\mathcal{L}_a^{(\rho)}$ since the reweighting scalars are nonzero, so $H_n^{(\rho)}(a)$ differs from $H_n(a)$ by a nonzero determinant of diagonal reweighting factors), so they transfer verbatim. $\square$

Lemma 9 (Neighboring-pair series) $L(a, t) = t G(a, x)$ for $x = t^\alpha$, where $G(a, x) = \sum_{j \geq 0} g_j(a) x^j \in \mathbb{C}[a][[x]]$ and, with $c_0 := 0$,

$$g_j(a) = \theta \frac{c_j(a)}{j\alpha + 1} + V_0 c_{j+1}(a) \frac{\Gamma((j+1)\alpha + 1)}{\Gamma(j\alpha + 2)}.$$

Proof Term by term, $\int_0^t s^{k\alpha} ds = t^{k\alpha+1}/(k\alpha + 1)$, so $\theta \int_0^t h ds = t \sum_{k \geq 1} \theta c_k(a) x^k / (k\alpha + 1)$. By the term-by-term Riemann–Liouville rule $I^\rho s^{k\alpha} = \frac{\Gamma(k\alpha+1)}{\Gamma(k\alpha+1+\rho)} t^{k\alpha+\rho}$ with $\rho = 1 - \alpha$: $V_0 I^{1-\alpha} h(t, a) = V_0 \sum_{k \geq 1} c_k(a) \frac{\Gamma(k\alpha+1)}{\Gamma((k-1)\alpha+2)} t^{(k-1)\alpha+1} = t \sum_{j \geq 0} V_0 c_{j+1}(a) \frac{\Gamma((j+1)\alpha+1)}{\Gamma(j\alpha+2)} x^j$ under $j = k - 1$. Add the two series and collect the coefficient of $t x^j$. $\square$

Lemma 10 (Local radius of convergence, uniform near $a = 0$) For every $\epsilon > 0$ there is $\rho_0(\epsilon) > 0$ such that $\sum_{k \geq 1} |c_k(a)| \rho^k < \infty$ for every $|a| \leq \epsilon$ and every $\rho < \rho_0(\epsilon)$; consequently $h(\cdot, a)$ and $G(a, \cdot)$ are holomorphic on $|x| < \rho_0(\epsilon)$, uniformly for $|a| \leq \epsilon$.

Proof [Majorant series] Fix $\epsilon > 0$ and set $q := \sup_{|a| \leq \epsilon} |Q(a)|$, $r := \sup_{|a| \leq \epsilon} |R(a)|$, $p := \sup_{|a| \leq \epsilon} |P(a)|$, all finite since P, Q, R are polynomials. The Gamma-ratio weights $\gamma_k := \Gamma(k\alpha + 1)/\Gamma((k+1)\alpha + 1)$ satisfy $\gamma_k \leq \gamma^* := \sup_{k \geq 1} \gamma_k < \infty$ (by Stirling, $\gamma_k = O(k^{-\alpha})$, and γ_k is a continuous, hence bounded, function of k on the finite range where it is not already decreasing). Set $u_1 := \gamma'_0 p$, $\gamma'_0 := 1/\Gamma(\alpha + 1)$, and $u_{k+1} := \gamma^* (q u_k + r \sum_{i+j=k} u_i u_j)$. Termwise application of the triangle inequality to the Müntz–Tau recurrence, together with $\gamma_k \leq \gamma^*$, gives $|c_k(a)| \leq u_k$ for all $|a| \leq \epsilon$ and all $k \geq 1$, by induction on k . The generating function $U(z) := \sum_{k \geq 1} u_k z^k$ satisfies, summing the recurrence against z^{k+1} ,

$$U - \gamma'_0 p z = \gamma^* q z U + \gamma^* r z U^2, \quad \text{i.e.} \quad \gamma^* r z U^2 + (\gamma^* q z - 1) U + \gamma'_0 p z = 0,$$

a quadratic in U with coefficients holomorphic in z and discriminant $\Delta(z) := (\gamma^* q z - 1)^2 - 4 \gamma^* \gamma'_0 p r z^2$ satisfying $\Delta(0) = 1 \neq 0$. The branch of U with $U(0) = 0$ is holomorphic on the disk $|z| < \rho_0(\epsilon)$, $\rho_0(\epsilon)$ the distance from 0 to the nearest zero of Δ (a nonzero, hence positive, quantity), so $\sum u_k z^k$ converges absolutely there; take any $\rho < \rho_0(\epsilon)$. $\square$

Lemma 11 (Denominator clearing) $H_n(a) \Pi(n, X) \in \mathbb{C}[a][X]$ and $H_n(a) \Pi^{(1)}(n, X) \in \mathbb{C}[a][X]$ for every n .

Proof By Heine’s determinantal formula (Definition 5), $H_n(a) \Pi(n, X)$ is, up to the sign of a row permutation, the determinant of a matrix whose entries are the $m(i, a) = c_{i+1}(a) \in \mathbb{C}[a]$ (Lemma 5) together with the monomials $1, X, \dots, X^n$; a determinant of polynomial entries is a polynomial, so $H_n(a) \Pi(n, X) \in \mathbb{C}[a][X]$. For $\Pi^{(1)}$: $\Pi^{(1)}(n, X) = \mathcal{L}_{a,Y}[(\Pi(n, X) - \Pi(n, Y))/(X - Y)]$, and $(p(X) - p(Y))/(X - Y) \in \mathbb{C}[X, Y]$ for any polynomial p ; substituting $H_n(a) \Pi(n, X) \in \mathbb{C}[a][X]$ for p and applying $\mathcal{L}_{a,Y}$ (which sends $Y^k \mapsto m(k, a) \in \mathbb{C}[a]$, $\mathbb{C}$ -linearly) termwise to the resulting polynomial in (a, X, Y) gives $H_n(a) \Pi^{(1)}(n, X) \in \mathbb{C}[a][X]$. $\square$

Lemma 12 (Rational Padé data for G) For each M there is a rational function $G_M(a, x) = A_M(a, x)/B_M(a, x)$ with $A_M, B_M \in \mathbb{C}[a][x]$, $\deg_x A_M, \deg_x B_M \leq 2M$, such that $G_M(a, \cdot)$ agrees with $G(a, \cdot)$ to order x^{2M} at $x = 0$.

Proof Apply Theorem 18 and Theorem 19 once to the moment functional $\mathcal{L}_a^{(1)}$ of $Y(t, a) = \theta \int_0^t h \, ds$ and once to $\mathcal{L}_a^{(1-\alpha)}$ of $V_0 I^{1-\alpha} h$ (Theorem 22), producing diagonal $[M/M]$ Padé pairs $\hat{P}^{(1)}/\hat{Q}^{(1)}$ and $\hat{P}^{(2)}/\hat{Q}^{(2)}$, each matching its series to order x^{2M+2} (Theorem 19). By Lemma 11 applied to each reweighted functional, $\tilde{P}^{(i)} := H_M^{(i)}(a)\hat{P}^{(i)}$ and $\tilde{Q}^{(i)} := H_M^{(i)}(a)\hat{Q}^{(i)}$ lie in $\mathbb{C}[a][x]$ ($H_M^{(i)}$ the Hankel determinant of $\mathcal{L}_a^{(i)}$), with $\tilde{P}^{(i)}/\tilde{Q}^{(i)} = \hat{P}^{(i)}/\hat{Q}^{(i)}$. Set

$$A_M := \tilde{P}^{(1)}\tilde{Q}^{(2)} + \tilde{P}^{(2)}\tilde{Q}^{(1)}, \quad B_M := \tilde{Q}^{(1)}\tilde{Q}^{(2)} \in \mathbb{C}[a][x].$$

Then $A_M/B_M = \hat{P}^{(1)}/\hat{Q}^{(1)} + \hat{P}^{(2)}/\hat{Q}^{(2)}$ agrees with $Y + I^{1-\alpha}h$, hence (weighting by θ, V_0 and dividing by t , i.e. Lemma 9) with $G(a, \cdot)$, to order x^{2M} (the lower of the two orders x^{2M+2} contributes after the division implicit in forming G from L); the degree bound $\deg_x A_M, \deg_x B_M \leq 2M$ is immediate from $\deg_x \tilde{P}^{(i)}, \deg_x \tilde{Q}^{(i)} \leq M$. $\square$

Define $\varphi_M(a, t) := \exp(tG_M(a, t^\alpha))$. For fixed a and t , φ_M is the exponential of a rational function of x : exactly the object of this paper.

Theorem 23 (Local data of φ_M in the coefficient algebra) *Fix a and M , and let $\xi_1, \dots, \xi_r$ be the distinct zeros of $B_M(a, \cdot)$ in $\mathbb{C}$, with multiplicities $\mu_1, \dots, \mu_r$, none lying on $[0, x^*]$ where $x^* = t^\alpha$ for the maturity range of interest (guaranteed for a real by Theorems 20–21 applied through Theorem 22 and Lemma 12, for all $M \geq M_0(a, x^*)$). Write the exact partial fraction decomposition*

$$tG_M(a, x) = \pi(x) + \sum_{i=1}^r \sum_{j=1}^{\mu_i} \frac{c_j^{(i)}}{(x - \xi_i)^j}, \quad \pi \in \mathbb{C}[x], \deg \pi \leq \deg A_M - \deg B_M + (\text{polynomial part}),$$

all data computable exactly from (A_M, B_M) by division and residue extraction in $\mathbb{C}[a](x)$. Then:

1. $\varphi_M(a, \cdot)$ is analytic on a neighborhood of $[0, x^*]$ in the x -plane and is evaluated there by direct composition $\exp \circ (tA_M/B_M)$; no expansion is needed on the ray and no essential singularity is approached.
2. Around each ξ_i , with $w = x - \xi_i$ and $H_i := tG_M - \sum_j c_j^{(i)} w^{-j}$ analytic on $|w| < r_i$ ($r_i = \min_{i' \neq i} |\xi_{i'} - \xi_i|$),

$$\varphi_M = \left(\sum_{N \geq 0} \Lambda_N(c^{(i)}) w^{-N} \right) \left(\sum_{l \geq 0} b_l^{(i)} w^l \right) \quad (6)$$

$$A_n^{(i)} = \sum_{k \geq \max(0, -n)} \Lambda_k(c^{(i)}) b_{n+k}^{(i)} \quad (7)$$

absolutely convergent Laurent data on $0 < |w| < r_i$, with truncation error controlled by Proposition 4 and with each coefficient sequence P -recursive via (3) applied to $R = tG_M$.

3. All quantities in (i)–(ii) are algebraic expressions in the Chebyshev–Wheeler output and hence carry rigorous arb enclosures when evaluated in ball arithmetic.

Proof (i) is the zero-freeness of B_M on the ray plus continuity of $\exp$. (ii) is Theorem 11 and Proposition 4 applied to the rational function $R(x) = tG_M(a, x)$ at the pole ξ_i (the polynomial part π is absorbed into H_i), together with Theorem 15. (iii) is by inspection: partial fractions, Λ -recurrence (1), and Taylor coefficients of e^{H_i} (themselves obtainable from (3)) are finite exact recurrences. $\square$

9.2 Price convergence without contour deformation

We deliberately do *not* deform any contour across poles of B_M : the poles belong to the approximant, not to φ , and residue exchange across them is not justified. Instead the pricing contour is held fixed inside the strip of analyticity of the exact characteristic function, and convergence is proved by domination.

Definition 9 (Lewis-type pricing functional) Fix maturity $T > 0$, $x^* = T^\alpha$, log-strike k , and a horizontal contour $\Im a = \eta_0$ interior to the moment-analyticity strip of $\varphi(\cdot, T)$. A Lewis-type functional is

$$P[\psi] = c_0 + \int_{\mathbb{R}} K(u) \psi(u + i\eta_0) du, \quad |K(u)| \leq \frac{C_K}{1 + u^2},$$

with c_0, K depending only on $(S_0, K, r, T, k, \eta_0)$; the call price is $P[\varphi(\cdot, T)]$ for the standard choices of (c_0, K, η_0) (Lewis [17]; Carr–Madan [4]; Gil-Pelaez [12]).

Theorem 24 (Convergence of Padé prices) Assume:

1. for every $u \in \mathbb{R}$, $B_M(u + i\eta_0, \cdot)$ is zero-free on $[0, x^*]$ for all $M \geq M_0(u)$, and $\varphi_M(u + i\eta_0, T) \rightarrow \varphi(u + i\eta_0, T)$ as $M \rightarrow \infty$;
2. there exists $\Phi \in L^1\left(\mathbb{R}, \frac{du}{1+u^2}\right)$ and M_1 with $|\varphi_M(u + i\eta_0, T)| \leq \Phi(u)$ for all $M \geq M_1$ and a.e. u .

Then $P[\varphi_M(\cdot, T)] \rightarrow P[\varphi(\cdot, T)]$: Padé prices converge to exact prices. Moreover (H1) holds pointwise on the contour when $\eta_0 = 0$ (the case used in §9.3) as a consequence of Theorems 20–21 applied through Theorem 22 and Lemma 12 at each fixed real $a = u$ (convergence of $G_M(a, \cdot) \rightarrow G(a, \cdot)$ on a neighborhood of $[0, x^*]$, hence of the exponentials); for $\eta_0 \neq 0$ the same conclusion holds wherever $\Im a = \eta_0$ avoids the (countable, per Lemma 6) discriminant locus, but pole-freeness there is only the generic Stahl statement, not the sharpened real-axis one, so the only unproved input in general is the domination (H2).

Proof For $M \geq \max(M_0(u), M_1)$ the integrand $K(u)\varphi_M(u + i\eta_0, T)$ converges pointwise to $K(u)\varphi(u + i\eta_0, T)$ by (H1) and continuity of nothing further (the integrand is the object itself), and is dominated by $C_K\Phi(u)/(1 + u^2) \in L^1$ by (H2). Apply dominated convergence. The claim about (H1): for fixed a on the contour, Stahl convergence in capacity together with zero-freeness of B_M on a fixed neighborhood $\Omega \supset [0, x^*]$ upgrades, by the standard capacity-to-uniform argument on compacts avoiding the poles (Stahl [24, Thm. 1 and §6]), to uniform convergence $G_M \rightarrow G$ on $[0, x^*]$; then $\varphi_M = \exp(TG_M(a, T^\alpha)) \rightarrow \exp(TG(a, T^\alpha)) = \varphi$. $\square$

Remark 8 (Status of (H2)) (H2) is the precise statement of what remains open on the analytic side. It is implied by a uniform-in- M , integrable-in- u bound on $\Re T G_M(u + i\eta_0, T^\alpha)$ from above. The leading term $g_0(a) = V_0 c_1(a)/(\alpha+1) = V_0 P(a)/((\alpha+1)\Gamma(\alpha+1))$ of Lemma 9 is, since $P(a) = \frac{1}{2}(-a^2 - ia)$, quadratic in a with negative leading coefficient on the real axis; this is the source of the expected Gaussian-type decay of $|\varphi_M(u, T)|$ for large real u , which is the classical Heston-family mechanism making such Fourier-pricing integrals converge in practice. Making this asymptotic dominate the M -dependent corrections $g_j(a)$, $j \geq 1$, uniformly in M is exactly hypothesis (H2); it is not established here. A proof of (H2) is Open Problem 1; Theorem 24 isolates it as the exact remaining gap, replacing the invalid contour-deformation route.

9.3 Explicit call and put prices via Gil–Pelaez inversion

Fix maturity T , spot S_0 , strike K , rate r , and set $k := \log(K/S_0) - rT$, so that $X_T = \log(S_T/S_0) - rT$ (the driftless discounted log-price of §9) has characteristic function $E[e^{iuX_T}] = \varphi_T(u)$ for real u .

Lemma 13 (Driftless normalization) $P(-i) = 0$, hence $h(t, -i) \equiv 0$, $L(-i, t) \equiv 0$ for all t , and $\varphi_T(-i) = E[e^{X_T}] = 1$.

Proof $P(-i) = \frac{1}{2}(-(-i)^2 - i(-i)) = \frac{1}{2}(1 - 1) = 0$. The constant function $h \equiv 0$ then satisfies $D^\alpha h = P(-i) + Q(-i)h + Rh^2 = 0 + 0 + 0$ with $I^{1-\alpha}h(0, -i) = 0$; by the uniqueness half of Lemma 4 (applied at $a = -i$), $h(\cdot, -i) \equiv 0$ is the solution. Hence $L(-i, t) = \theta \int_0^t 0 \, ds + V_0 I^{1-\alpha} 0 = 0$ for every t , so $\varphi_T(-i) = e^{L(-i, T)} = e^0 = 1$. $\square$

Theorem 25 (Gil–Pelaez call/put formula) Suppose φ_T extends to a function on the strip $-1 \leq \Im a \leq 0$ that is holomorphic in the interior and continuous on the boundary lines $\Im a \in \{0, -1\}$. Then, with

$$P_2 := \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \Re \left[\frac{e^{-iuk} \varphi_T(u)}{iu} \right] du, \quad P_1 := \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \Re \left[\frac{e^{-iuk} \varphi_T(u-i)}{iu} \right] du,$$

(both integrands extend continuously across $u = 0$: writing $\psi(u) \in \{\varphi_T(u), \varphi_T(u-i)\}$, $\psi(0) = 1$, so $e^{-iuk}\psi(u)/(iu) = -i/u + [e^{-iuk}\psi(u) - 1]/(iu)$, and the bracketed numerator is $O(u)$ as $u \rightarrow 0$ (it vanishes at $u = 0$ and is differentiable there), so the second term is bounded near 0; the divergent first term $-i/u$ is purely imaginary and is annihilated by $\Re[\cdot]$), if in addition $u \mapsto \Re[e^{-iuk}\psi(u)/(iu)]$ is absolutely integrable on $(0, \infty)$ for $\psi \in \{\varphi_T, \varphi_T(\cdot - i)\}$, the exact European prices are

$$C(K, T) = S_0 P_1 - K e^{-rT} P_2, \quad \text{Put}(K, T) = K e^{-rT} (1 - P_2) - S_0 (1 - P_1).$$

Proof By the Gil–Pelaez inversion formula (Gil–Pelaez [12]) applied to the real random variable X_T with characteristic function φ_T ,

$$\Pr(X_T > k) = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \Re \left[\frac{e^{-iuk} \varphi_T(u)}{iu} \right] du = P_2.$$

By Lemma 13, $E[e^{X_T}] = 1 > 0$, so the Radon–Nikodym derivative e^{X_T} (nonnegative, expectation 1) defines a new probability measure via $\tilde{E}[\cdot] := E[e^{X_T} \cdot]$. Under $\tilde{E}$, X_T has characteristic function $\tilde{E}[e^{iuX_T}] = E[e^{X_T} e^{iuX_T}] = E[e^{i(u-1)X_T}] = \varphi_T(u-1)$ (the middle equality is $e^{X_T} e^{iuX_T} = e^{(1+iu)X_T} = e^{i(u-1)X_T}$, and φ_T analytically continues to $a = u-1$ by hypothesis). Gil–Pelaez applied to X_T under $\tilde{E}$ gives $\Pr(X_T > k) := \tilde{E}[\mathbf{1}_{X_T > k}] = P_1$. By definition of $\tilde{E}$, $\tilde{\Pr}(X_T > k) = E[e^{X_T} \mathbf{1}_{X_T > k}]$. Since $e^k = K e^{-rT}/S_0$,

$$E[(e^{X_T} - e^k)^+] = E[e^{X_T} \mathbf{1}_{X_T > k}] - e^k \Pr(X_T > k) = P_1 - e^k P_2.$$

Now $S_T = S_0 e^{X_T + rT}$, so $(S_T - K)^+ = S_0 e^{rT} (e^{X_T} - e^k)^+$, and

$$C(K, T) = e^{-rT} E[(S_T - K)^+] = S_0 E[(e^{X_T} - e^k)^+] = S_0 (P_1 - e^k P_2) = S_0 P_1 - K e^{-rT} P_2.$$

Put–call parity: $C - \text{Put} = e^{-rT} E[(S_T - K)^+ - (K - S_T)^+] = e^{-rT} E[S_T - K] = e^{-rT} (S_0 e^{rT} - K) = S_0 - K e^{-rT}$ (using $E[S_T] = S_0 e^{rT} E[e^{X_T}] = S_0 e^{rT}$ from Lemma 13), so $\text{Put} = C - S_0 + K e^{-rT} = S_0 (P_1 - 1) + K e^{-rT} (1 - P_2) = K e^{-rT} (1 - P_2) - S_0 (1 - P_1)$. $\square$

Corollary 3 (Padé call/put with certified real-axis evaluation) *For every M and $u \in \mathbb{R} \setminus \{0\}$, if $B_M(u, T^\alpha) \neq 0$ then $\varphi_M(u, T) = \exp(T A_M(u, T^\alpha)/B_M(u, T^\alpha))$ is defined by direct evaluation, with rigorous ball-arithmetic enclosures by part (iii) of Theorem 23; by Theorem 20 applied at $a = u$ (real), $B_M(u, T^\alpha) \neq 0$ for all $M \geq M_0(u, T^\alpha)$. Define $P_2^{[M]}, P_1^{[M]}, C_M, \text{Put}_M$ by substituting φ_M for φ_T throughout Theorem 25 (on the two contours $\Im a \in \{0, -1\}$). If $\varphi_M \rightarrow \varphi_T$ pointwise on these contours (Theorem 24, first part of (H1), which holds unconditionally at $\eta_0 = 0$ by Theorem 24's refinement above) and is dominated there in the sense of the analogue of (H2) for the kernel $K(u) = (2\pi i u)^{-1} e^{-iuk}$, then $C_M \rightarrow C$ and $\text{Put}_M \rightarrow \text{Put}$.*

Proof Zero-freeness of $B_M(u, \cdot)$ at $x = T^\alpha$ for real u : by Lemma 12, $B_M(u, x) = \tilde{Q}^{(1)}(u, x) \tilde{Q}^{(2)}(u, x)$, and each $\tilde{Q}^{(i)}$ is, up to the nonzero scalar $H_M^{(i)}(u)$ (Lemma 11), the reciprocal-flipped denominator $\hat{Q}^{(i)}(M, \cdot)$ of a diagonal Padé approximant covered by Theorem 22; by Theorem 20 (applied to the corresponding reweighted moment functional at the real parameter $a = u$) and Theorem 21, $\hat{Q}^{(i)}(M, \cdot)$ has no zero on $[0, T^\alpha]$ for M at least some $M_0^{(i)}(u, T^\alpha)$; take $M_0 := \max(M_0^{(1)}, M_0^{(2)})$. Well-definedness and the ball-arithmetic enclosure are then Theorem 23(iii) applied at $a = u$ with the empty pole set on $[0, T^\alpha]$ (direct composition, part (i)). The convergence statement is Theorem 25 applied with φ_M in place of φ_T in $P_1^{[M]}, P_2^{[M]}$, combined with dominated convergence on each of the two contours exactly as in the proof of Theorem 24 (the kernel $K(u) = (2\pi i u)^{-1} e^{-iuk}$, extended continuously at $u = 0$ as in Theorem 25, plays the role of the abstract kernel of Definition 9 there). $\square$

Proposition 7 (Cumulants in the algebra) *Every $c_k(a)$, and hence every $g_j(a)$, is a polynomial in a (Lemma 5, Lemma 9), and for $0 \leq t < \rho_0(\epsilon)^{1/\alpha}$, uniformly for $|a| \leq \epsilon$ (Lemma 10), the cumulants of X_T are the absolutely convergent series*

$$\kappa_n(t) = \partial_a^n L(a, t)|_{a=0} = t \sum_{j \geq 0} (\partial_a^n g_j)(0) t^{\alpha j},$$

with each $(\partial_a^n g_j)(0)$ a finite exact expression obtained by Leibniz-differentiating the Müntz–Tau recurrence of Lemma 4 in a . Consequently the cumulant data feeding any transform- or expansion-based pricing representation lies in $\mathbb{C}$ -linear combinations of the same neighboring-pair objects, computed without approximation.

Proof Polynomiality of $c_k(a)$ is Lemma 5; $g_j(a)$ is then polynomial in a by its explicit formula in Lemma 9 (a $\mathbb{C}$ -linear combination, with constant scalar coefficients, of $c_j(a)$ and $c_{j+1}(a)$). For $|a| \leq \epsilon$ and $t < \rho_0(\epsilon)^{1/\alpha}$, $x = t^\alpha < \rho_0(\epsilon)$, so $\sum_j |g_j(a)| x^j < \infty$ uniformly in a by Lemma 10 together with the explicit formula for g_j (each term of g_j is a bounded scalar times c_j or c_{j+1} , whose sums $\sum |c_j(a)| \rho^j$ are already controlled there); a locally uniformly convergent series of polynomials in a may be differentiated termwise in a any number of times (Weierstrass), giving the display. Each $\partial_a^n g_j(0)$ is, by Lemma 9, the same $\mathbb{C}$ -linear combination (with the same constant scalars $\theta/(j\alpha + 1)$, $V_0 \Gamma((j+1)\alpha + 1)/\Gamma(j\alpha + 2)$) of $\partial_a^n c_j(0)$ and $\partial_a^n c_{j+1}(0)$; these obey the recurrence obtained by applying the Leibniz rule to the Müntz–Tau recurrence of Lemma 4 — $\partial_a^n c_{k+1} = \frac{\Gamma(k\alpha+1)}{\Gamma((k+1)\alpha+1)} \sum_{i=0}^n \binom{n}{i} \left((\partial_a^i Q)(\partial_a^{n-i} c_k) + (\partial_a^i R) \sum_{p+q=k} \sum_{l=0}^{n-i} \binom{n-i}{l} (\partial_a^l c_p)(\partial_a^{n-i-l} c_q) \right)$ — closed and finite at every order because P, Q, R are polynomials in a (so all a -derivatives beyond their degree vanish). $\square$

Algorithm 1 (Coefficient-algebra evaluation layer for arb4j) *Input:* $(A_M, B_M) \in \mathbb{C}[a][x]^2$ from the Chebyshev–Wheeler pipeline; contour height η_0 ; maturity T ; working precision; truncation bandwidth K .

1. For each quadrature node $a = u + i\eta_0$: evaluate $B_M(a, x)$ on $[0, T^\alpha]$ in ball arithmetic; certify zero-freeness (interval Descartes/argument principle). Cost $O(M)$ per node.
2. Price by fixed-contour quadrature of $K(u) \exp(T A_M/B_M)$ (Theorem 24); rigorous tails from the kernel bound. Cost: nodes $\times O(M)$.
3. (Diagnostics / analytic continuation off the ray.) For each required pole ξ_i : partial fractions to get $c^{(i)}$ ($O(M^2)$ exact ops); Λ -sequence by (1) ($O(mK)$ ops); Taylor data $b_l^{(i)}$ of e^{H_i} by (3) ($O((d_U + d_V)K)$ ops); assemble $A_n^{(i)}$ with certified tail via Proposition 4.
4. Cumulant channel: $\partial_a^n c_k(0)$ by the differentiated Gamma-ratio recurrence ($O(nM)$ per order), then κ_n by Proposition 7.

All steps are exact recurrences or ball-arithmetic evaluations with explicit error terms; no Hankel matrix is ever formed or inverted (Theorem 18), matching the stability profile established there.

10 Open problems

Problem 1 (Domination) Prove hypothesis (H2) of Theorem 24 (equivalently, its analogue in Corollary 3 for the undamped Gil–Pelaez kernel of §9.3): a uniform-in- M integrable majorant for $|\varphi_M|$ on the pricing contour, e.g. via a uniform-in- a version of the Stahl rate of Theorem 21 combined with the quadratic leading-order decay of $g_0(a)$ identified in Remark 8. What is *not* open is real-axis pole-freeness of B_M itself (Theorem 20, used directly in Corollary 3); the remaining gap is purely the uniform-in- M decay rate of $|\varphi_M|$.

Problem 2 (Resurgent structure) e^Π is the exponential factor of the Hukuhara–Levelt–Turrittin normal form of the rank-one connection $d - dR$ at the irregular singularity a ([26, Ch. 3]); this much is classical. Construct the action of the alien derivations of Écalle on the $\mathfrak{B}_m$ -module of Theorem 11 — the natural conjecture, suggested by Lemma 3, is that singular directions act through the lowering operators ∂_{c_j} — and determine the Stokes data of φ_M in terms of the $c^{(i)}$ (framework: Mitschi–Sauzin [21]; Costin [6]). No such statement is asserted here as a theorem.

Problem 3 (Exponential periods) $e^R dz$ is an exponential-period integrand; the category of exponential motives (Fresán–Jossen [11]) attaches cohomological invariants to $(\mathbb{A}^1, R)$. Position $\mathfrak{B}_m$ relative to the de Rham realization of the rank-one exponential connection: is the coefficient module of Theorem 11 a concrete model of a known object there?

Problem 4 (q -lift) Lift (Λ_N) along the Hall-algebra isomorphism of Remark 3 to the Hall–Littlewood level: identify the q -analogue of the universal recurrence and its analytic meaning, if any, for q -deformed exponentials of principal parts.

Problem 5 (Non-reduction) Prove (or refute) that for generic (c_1, c_2, c_3) the sequence $(\Lambda_N(c_1, c_2, c_3))_N$ is not expressible as $(\rho^N F(\sigma N + \tau))_N$ for any single classical hypergeometric-type function F of one variable — i.e. make Remark 5 a theorem, plausibly via differential Galois groups of the order-3 holonomic equation from (1).

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